

Carbon-plate insole use and self-reported injury prevalence among Italian competitive athletics athletes: a cross-sectional, Bayesian multilevel analysis of a two-phase recruited cohort

Anonymized Author 1

Anonymized Author 2

Abstract

Objective: To describe carbon-plate insole use during preparation and competition among competitive Italian athletics athletes, estimate its cross-sectional, period-matched association with self-reported current-season injury, and examine whether that association was better characterized by relative carbon share or absolute carbon-use frequency.

Design: Cross-sectional survey with two-phase (voluntary, then targeted-recall) recruitment, poststratified to national age–sex population strata; reported in line with STROBE and the STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) extension.

Participants: FIDAL-affiliated athletics athletes (Under 20, Under 23, Senior categories) recruited nationally in 2026. Of 355 eligible, complete respondents, 352 (704 athlete-period records) formed the balanced analysis cohort.

Exposure: Carbon share (the proportion of reported weekly training-type frequency delivered in carbon-plated spikes) and absolute carbon-frequency score (the summed carbon-frequency items) across five training domains, calculated separately for preparation and competition.

Outcome: Self-reported injury occurring in the preparation period and/or the competition period of the current season.

Statistical methods: Bayesian multilevel logistic models used period-specific intercepts, an athlete-level random intercept, and adjustment for demographic, anthropometric, period-matched training-frequency, and discipline covariates. The primary model represented carbon share among users with a continuous, data-driven curve fitted by a Hilbert-space Gaussian process, letting the amount of curvature in the dose–response relationship be estimated rather than fixed in advance. Two matched linear models then evaluated competing relative-use (carbon share) and absolute-use (carbon-frequency score) hypotheses on a common among-user standard-deviation scale. Estimands included adjusted prevalence contrasts and conditional absolute-risk differences for a fixed reference profile.

Results: Carbon-plated spikes accounted for a mean 35.7% of reported preparation training and 47.7% of reported competition training. Adoption showed no clear association with injury in either period. The primary curve showed adjusted competition-period injury prevalence climbing steadily with carbon share; moving from the 25th to 75th percentile of competition-period carbon share was associated with a 13.2-percentage-point higher adjusted injury prevalence (95% CrI –0.2 to 29.1; $P_{\text{dir}} = 96.9\%$), the strongest of the four primary contrasts and the only one clearing the pre-specified practical-relevance threshold. In the matched linear comparison, a one-standard-deviation increase meant 24.1 percentage points of carbon share (37.3% to 61.4%) or 3.36 units of the summed ordinal carbon-frequency score (2.83 to 6.19). Among competition users, these increments corresponded to risk differences of 7.9 points for share (95% CrI 1.4 to 14.4; $P_{\text{dir}} = 99.0\%$) and 8.3 points for absolute score (95% CrI 1.6 to 15.9; $P_{\text{dir}} = 99.2\%$); preparation-period estimates were 3.6 points (95% CrI –3.9 to 11.2; $P_{\text{dir}} = 82.8\%$) and 1.5 points (95% CrI –6.9 to 9.4; $P_{\text{dir}} = 63.2\%$), respectively.

Conclusion: Among competition users, greater reported carbon exposure was associated with higher adjusted injury prevalence whether expressed as relative share or absolute frequency score; these data did not identify one scale as the better explanation. Adoption and preparation-period exposure were not clearly associated with injury. The cross-sectional, retrospective design does not support causal interpretation.

Keywords: athletics; track and field; carbon-fiber plate; advanced footwear technology; injury surveillance; Bayesian multilevel model; STROBE-SIIS.

1 Introduction

Athletics (track and field) carries a substantial burden of health problems at the competitive tiers of participation [McKay et al., 2022]. A systematic review and meta-analysis of 216 studies in Olympic athletics disciplines reported a mean injured-athlete prevalence of 50.4% (range 2.4%–93.6%) and a mean incidence of 135.6 injuries per 1000 registered athletes, with injury patterns differing markedly by discipline group, tissue type, and period of the season [Edouard et al., 2020]. Roughly half of injuries in youth athletics academies were reported in the winter preparation period [Martínez-Silván et al., 2021], motivating separate description of preparation- and competition-period injury rather than a single season-level estimate.

How injury is registered materially changes the burden observed. Surveillance restricted to absence, complete sports cessation, or medical attention can under-ascertain overuse problems because athletes may continue training and competing despite symptoms and impaired function [Clarsen et al., 2013]. The Oslo Sports Trauma Research Center Questionnaire on Health Problems (OSTRC-H) therefore complements time-loss registration with direct athlete reports across four impact domains: sports participation, training volume, performance, and symptoms [Clarsen et al., 2014, 2020]. These domains provide an athlete-reported measure of sport-specific health burden; they should not be interpreted as a generic quality-of-life scale.

Since 2020–2021, competitive athletics has undergone a rapid equipment transition with the introduction of “advanced footwear technology” (AFT) spikes: a rigid, longitudinally curved carbon-fiber plate embedded in a high-resilience foam midsole, first validated in road racing (Nike Vaporfly, Breaking2 project) and subsequently extended to track spikes for sprint and middle-distance events [Healey et al., 2022]. The stated biomechanical rationale is that plate-driven longitudinal bending stiffness reduces energy dissipation at the metatarsophalangeal joints and shifts the center of pressure anteriorly, increasing horizontal propulsive force [Willwacher et al., 2016]. Comparative analyses of world-leading sprint performance before and after the introduction of AFT spikes have documented step changes in top-100 performance lists coincident with the technology’s diffusion [Mason et al., 2023].

The same mechanical changes that plausibly improve performance also alter loading at the ankle and forefoot. A recent kinetic comparison of traditional and carbon-plated sprint spikes in elite sprinters found materially higher ankle joint contact force and higher peak plantarflexor muscle force in the carbon-plated condition, with pronounced inter-individual variability [Gerlach et al., 2025]. Case reports and narrative reviews have raised the possibility of increased Achilles tendon load and of altered forces at the navicular and cuneiform bones, both recognized high-risk sites for bone stress injury in runners [Tenforde et al., 2023]. Evidence on ground-reaction-force-mediated risk is mixed, and no consensus exists on whether increased longitudinal bending stiffness is protective, neutral, or hazardous once inter-individual differences in plantarflexor strength are taken into account [Willwacher et al., 2016, Gerlach et al., 2025].

Despite this plausible mechanistic pathway, direct epidemiological evidence on carbon-plate in-

sole use and injury in competitive athletics is essentially absent: available data are limited to isolated case reports and small mechanistic cohorts, none of which were designed to estimate population-level injury associations [Tenforde et al., 2023, Gerlach et al., 2025]. Established risk factors for running-related injury more broadly include age, body mass index, weekly training frequency, footwear, years of sporting experience, and prior injury [Taunton et al., 2003], all of which needed to be accounted for before any exposure–outcome association attributable to carbon-plate use could be considered informative.

This study was designed to close part of this gap by describing carbon-plate insole use among a national sample of Italian competitive athletics athletes and by estimating the cross-sectional, period-matched association between reported carbon share and self-reported injury during the preparation and competition periods of a single season. Consistent with the Injury Definitions Concept Framework and the athletics-specific consensus statement on injury and illness surveillance [Timpka et al., 2014, 2015], injury was defined broadly as a physical complaint or observable tissue damage attributable to the transfer of energy experienced by an athlete during training or competition participation in athletics, irrespective of whether medical attention was sought or of the consequences for participation [Bahr et al.]. The primary hypothesis, formulated a priori, was that greater reported carbon share would be associated with higher adjusted injury prevalence; a secondary, exploratory aim was to describe whether carbon-plate adoption and share differed by age group and sex, and whether the exposure–outcome association was stable under population reweighting and alternative dose–response specifications. To clarify the scale of any observed association, a secondary matched-model comparison subsequently examined two explicit exposure hypotheses: whether injury tracked relative allocation to carbon-plated spikes (carbon share) or absolute reported carbon-use frequency after adjustment for total training frequency.

2 Methods

2.1 Study design and reporting

This is a retrospective, cross-sectional survey study of competitive athletics athletes affiliated with the Federazione Italiana di Atletica Leggera (FIDAL). Reporting follows the STROBE statement and its athletics-relevant STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) extension [von Elm et al., 2007, Bahr et al.]; the completed checklist, with the manuscript line number supporting each item, is given in Appendix A.

2.2 Study registration

This study was not registered in a public trial or study registry (for example, ClinicalTrials.gov or an equivalent observational-study registry). It was, however, prospectively specified through a complete study protocol, submitted in Italian and reviewed and approved by the Comitato Etico of the University of Turin (Section 2.3) before survey deployment. The protocol specified the study population, the two-phase recruitment and poststratification strategy, the exposure and outcome definitions, and the primary hypothesis (Section 1) in advance of data collection and of any outcome analysis. This ethics-committee-reviewed protocol is the prospective record of the study design; it is not, however, a substitute for public registration, and the absence of the latter is noted as a limit on independent, external verification of pre-specification (Section 4.4).

2.3 Setting and ethics

Data collection used a structured, self-administered online survey covering demographic, anthropometric, training, footwear, and injury-history items for the current competitive season, separately for the preparation and competition periods. Study data were collected and managed using REDCap electronic data capture tools hosted at the Department of Clinical and Biological Sciences, University of Turin [Harris et al., 2009, 2019]. REDCap (Research Electronic Data Capture) is a secure, web-based software platform designed to support data capture for research studies, providing (1) an intuitive interface for validated data capture, (2) audit trails for tracking data manipulation and export procedures, (3) automated export procedures for seamless data downloads to common statistical packages, and (4) procedures for data integration and interoperability with external sources.

The study was approved by the Comitato Etico of the University of Turin (protocol n. 0453001, approved 23 June 2026). Participation required electronic informed consent; no financial incentive was offered. Individual survey responses were stored under a study identifier disconnected from athlete names at the point of analysis; only aggregate, de-identified data and model outputs are reported here, in line with the data-privacy provisions of the Declaration of Helsinki.

2.4 Participants and two-phase recruitment

Eligible athletes were FIDAL-affiliated competitive athletes in the Under 20, Under 23, or Senior (*assoluta*) categories, across seven athletics disciplines with adequate cell support in the sample (sprint, middle distance, long distance, hurdles, horizontal jumps, pole vault, combined events; four further disciplines had zero respondents and were dropped from discipline-level adjustment; Appendix Table 14).

Recruitment proceeded in two phases to reduce the risk that a purely voluntary sample would over-represent already engaged or higher-exposure athletes. In Phase 1, the survey was distributed openly through FIDAL-affiliated clubs and athlete networks; enrollment was entirely voluntary. National population totals by age category, sex, and discipline group were estimated from FIDAL competition-registry history (Appendix Table 15). Study size was planned for precision of a population proportion rather than power for a specific exposure association. For each candidate sample size n , the worst-case count $x = \lfloor n/2 \rfloor$ was combined with a Beta(1,1) prior, giving a Beta($1+x$, $1+n-x$) posterior; the smallest n whose equal-tailed 95% credible interval had half-width no greater than 0.06 was selected. This rule gave $n = 264$ from the estimated eligible frame of 1328 athletes (19.9%); targets were allocated proportionally across the 42 age-by-sex-by-discipline strata using largest-remainder rounding (Appendix Table 16). The calculation established a minimum precision and population-coverage target, not representativeness by itself. After Phase 1 closed, strata that had not reached their target count were identified, and Phase 2 undertook targeted recall-direct outreach to athletes and clubs within under-represented age-by-sex-by-discipline strata-until every stratum met or approached its target or the recruitment window closed. Detailed recruitment-monitoring cells are retained privately to avoid disclosing small-cell participation patterns. This two-phase design does not produce a probability sample with known, non-zero, equal selection probabilities in every stratum; population coverage was supported by targeted recruitment, and residual over- and under-representation was corrected analytically (Section 2.9) rather than assumed away.

Of all survey respondents, 355 provided consent, met calculated eligibility, and completed the survey (`survey_atleta_it_complete` = 2). One athlete was excluded for incomplete injury-timing data, and two further athletes were excluded because a period (preparation or competition) had a zero total eligible training-frequency score, precluding a carbon-share denominator. The final balanced analysis cohort comprised 352 athletes contributing 704 athlete-period records (one prepa-

ration and one competition record per athlete; Table 3).

2.5 Exposure: carbon-plate insole use

For each period (preparation, competition) and each of five training domains—long endurance, middle endurance, lactate, maximum velocity, and technique—athletes reported an ordinal weekly training-type frequency (0, 1, 2, 3, 4, or 5-or-more sessions per week, with 5-or-more coded as 5) separately for sessions performed in carbon-plated spikes and sessions performed without them. *Carbon share* for a given period was defined as

$$\text{carbon share} = \frac{\sum \text{carbon-frequency items}}{\sum \text{carbon-frequency items} + \sum \text{non-carbon-frequency items}},$$

summed across the five domains. The corresponding *absolute carbon-frequency score* was the numerator alone, that is, the sum of the five carbon-frequency items. It is an ordinal frequency score rather than an exact count of unique sessions because responses were top-coded at five and a session could contribute to more than one reported training domain. Gym/strength conditioning (*palestra*) has no carbon-eligible equivalent and was retained only as a non-carbon adjustment covariate, not in the carbon-share denominator. Carbon share is therefore a composition of reported ordinal training-type frequency, not a directly measured training-time proportion or shoe-wear duration; this distinction is carried through to the interpretation of all exposure estimands. Exposure was decomposed into two components: *adoption* (any carbon use versus none, `carbon_frequency_sum` > 0) and, among users, continuous *dose*, expressed primarily as carbon share and additionally as the absolute carbon-frequency score.

2.6 Outcome: period-specific injury

The outcome was self-reported injury attributable to athletics training or competition participation during the current season, following the athletics-specific consensus definition [Bahr et al., Timpka et al., 2014]: any physical complaint or observable tissue damage resulting from the transfer of energy experienced by an athlete during participation, regardless of medical attention sought or time-loss consequence. Athletes reported the number of current-season injuries (0, 1, or 2) and, for each, whether onset occurred in the preparation period, the competition period, or both (coded 1, 2, or 3 respectively). Preparation-period injury was defined as any reported injury coded 1 or 3; competition-period injury as any reported injury coded 2 or 3. Athletes could therefore contribute a positive outcome to one or both periods.

For each reported injury, athletes also selected the anatomical location (hip, groin, thigh, lower leg, ankle, foot, or head/upper limb/trunk; multi-select) and, only for injuries localized to the lower limb, the injury type (joint sprain, tendon rupture, ligament injury, muscle strain, bone stress injury or stress fracture, tendinopathy, plantar fasciopathy, or other; multi-select, adapted from consensus-recommended categories [Bahr et al.]). These location and type items were not used as covariates or outcomes in the primary Bayesian model; they are reported descriptively (Section 3.5) to characterize the injury pattern in this cohort, following STROBE-SIIS item SIIS17.1. Injury severity (time-loss duration) was also collected but is outside the scope of the present analysis.

Athletes reporting at least one current-season injury were also presented with four items adapted from the updated OSTRC-H2, covering the injury’s impact on sports participation, training volume, sports performance, and symptoms [Clarsen et al., 2014, 2020]. The four ordered responses were scored 0, 8, 17, and 25, then summed to an adapted OSTRC severity score from 0 (no reported impact) to 100 (maximum reported impact). Because these items were administered once

retrospectively and only after a current-season injury was reported, the score is used solely as a descriptive athlete-reported impact measure; it is not a prospective weekly OSTRC prevalence or cumulative-burden estimate.

2.7 Covariates

Adjustment covariates were age (years), competition sex, height (cm), body mass (kg), years of athletics experience, injury in either of the two preceding seasons (binary), period-matched total training-type-frequency scores for each of the five domains and for *palestra*, and indicator variables for the seven discipline categories with non-zero support. These variables were selected a priori based on previously reported risk factors for running-related injury [Taunton et al., 2003] and on the requirement, specified in the STROBE-SIIS extension, to account for how athletes were categorized by sport, discipline, and competitive level [Bahr et al.]. Continuous covariates were standardized (zero mean, unit variance) before model fitting.

2.8 Statistical methods

All inference is Bayesian [Gelman et al., 2013]; no null-hypothesis-testing framework or p -values are reported. Results are summarized as posterior means, medians, standard deviations, and 95% credible intervals (CrI, equal-tailed 2.5th–97.5th percentiles). Every reported interval for a signed association contrast is accompanied by its posterior probability of direction, $P_{\text{dir}} = \max\{P(\Delta > 0), P(\Delta < 0)\}$ (with one replacing zero as the null for ratios): the posterior probability that the contrast lies on the same side of the null as its median estimate. This is a graded measure of directional certainty, not a p -value or a binary significance test, and it must be read together with the effect magnitude and CrI. P_{dir} is not reported for absolute prevalences, for which positivity is constrained by definition and therefore uninformative. We additionally report posterior probabilities that an association exceeds a pre-specified practically relevant threshold (5 percentage points for differences, a ratio of 1.10) or falls within a region of practical equivalence to no association (± 2 percentage points; ratio 0.95–1.05). Models were fit in Stan [Carpenter et al., 2017] via its R interface.

The adapted OSTRC scores were summarized by mutually exclusive injury-period groups (preparation only, competition only, or both periods) using medians and interquartile ranges. As a descriptive effect-size aid, standardized mean differences (SMD) compared preparation-only and both-period groups with the competition-only group, calculated as the mean-score difference divided by the square root of the average of the two group variances. No injured-versus-uninjured OSTRC comparison was made because the items were not administered to uninjured athletes.

2.8.1 Primary model

Each athlete contributed one Bernoulli outcome per period. A shared athlete-level random intercept linked the two outcomes, following a cluster-specific logistic formulation for correlated binary data [Neuhaus et al., 1991]. For athlete i in period p ,

$$Y_{ip} \sim \text{Bernoulli}(\pi_{ip}), \quad \text{logit}(\pi_{ip}) = \alpha_p + c_{ip} + a_{ip} + \delta_i + u_i, \quad u_i \sim \mathcal{N}(0, \sigma_u^2).$$

Here c_{ip} is the carbon-use contribution:

$$c_{ip} = \begin{cases} 0, & \text{for a non-user,} \\ \beta_p^{\text{any}} + f_p(\text{share}_{ip}), & \text{for a carbon user.} \end{cases}$$

The remaining terms have direct roles: α_p is the period-specific intercept, a_{ip} combines the athlete-level and period-matched covariates of Section 2.7, δ_i is the partially pooled discipline contribution, and u_i is the athlete random intercept. The function f_p is a period-specific curve over carbon share among users, given a Hilbert-space approximate Gaussian-process (HSGP) prior with a Matérn-5/2 kernel [Riutort-Mayol et al., 2023], fitted with a common basis and centered at the pooled median user share (0.50):

$$f_p(\text{share}_{ip}) = \sum_{m=1}^{40} \phi_m(\text{share}_{ip}) b_{p,m}, \quad b_{p,m} \sim \mathcal{N}\left(0, S\left(\sqrt{\lambda_m}; \alpha_p^{\text{gp}}, \rho_p^{\text{gp}}\right)\right),$$

where ϕ_m and λ_m are the eigenfunctions and eigenvalues of the Laplace operator on the centered, boundary-extended domain $[-0.75, 0.75]$ (a boundary factor of 1.5 times the largest centered observed user share), $S(\cdot)$ is the Matérn-5/2 spectral density, and α_p^{gp} and ρ_p^{gp} are period-specific marginal-standard-deviation and length-scale hyperparameters estimated from the data. This construction lets the posterior determine how much curvature the competition- and preparation-period dose-response relationships each support directly from the data, through α_p^{gp} and ρ_p^{gp} . 40 basis functions were confirmed sufficient by checking that the spectral density at the highest included frequency was negligible relative to the lowest across both hyperparameters' posterior distributions. Consequently, β_p^{any} compares a median-share user with a non-user on the conditional log-odds scale, while f_p describes dose among users. The athlete and discipline terms, and the HSGP basis coefficients $b_{p,m}$, were parameterized non-centered, with discipline coefficients partially pooled through a shared scale τ_{disc} . Priors were regularizing and weakly informative on the standardized log-odds scale: $\alpha_p \sim \mathcal{N}(\text{logit}(0.4), 1.5^2)$, $\beta_p^{\text{any}} \sim \mathcal{N}(0, 0.7^2)$, covariate coefficients $\mathcal{N}(0, 0.5^2)$, $\tau_{\text{disc}} \sim \text{Half-N}(0, 0.5^2)$, $\sigma_u \sim \text{Half-N}(0, 1^2)$, $\alpha_p^{\text{gp}} \sim \text{Half-N}(0, 0.7^2)$, and $\rho_p^{\text{gp}} \sim \text{Inv-Gamma}(2.94, 0.80)$, the last chosen so that its 1st and 99th prior percentiles fell at one-tenth and twice the observed centered-share range, discouraging length scales too short for the data to resolve while still permitting substantial curvature. A matched baseline model retained the identical outcome, covariate, discipline, and athlete-random-intercept structure but removed both carbon-use terms, providing a no-exposure comparator for predictive-validity assessment (Section 2.8.4).

Figure 1 summarizes the primary model's observed inputs, parameter hierarchy, and repeated athlete-period outcome structure.

2.8.2 Estimands and marginal standardization

Posterior draws were used to compute marginal (population-standardized) adjusted prevalences by averaging model predictions over the observed empirical distribution of covariates and discipline indicators within each period, and by integrating the athlete random intercept using 15-point Gauss-Hermite quadrature rather than conditioning on a fixed value of u_i . For each period, adjusted prevalence was computed at four exposure scenarios: non-user, and user at the empirical 25th (P25), 50th (P50), and 75th (P75) percentile of the observed among-user share distribution for that period. These are ranks in the observed exposure distribution, not fixed share values: P25, P50, and P75 corresponded to 28.6%, 42.1%, and 57.1% share in preparation and to 36.0%, 50.0%, and 71.4% share in competition. Two primary contrasts were derived: the *adoption* contrast (P50 user vs. non-user; prevalence difference and ratio) and the *dose* contrast (P75 vs. P25 among users; prevalence difference and ratio).

To provide a directly interpretable absolute-risk presentation without fitting an additional model, the same posterior draws were also evaluated for one fixed reference profile. The profile was a male speed athlete aged 21 years, 176 cm tall, weighing 64 kg, with 10 years of athletics experience and

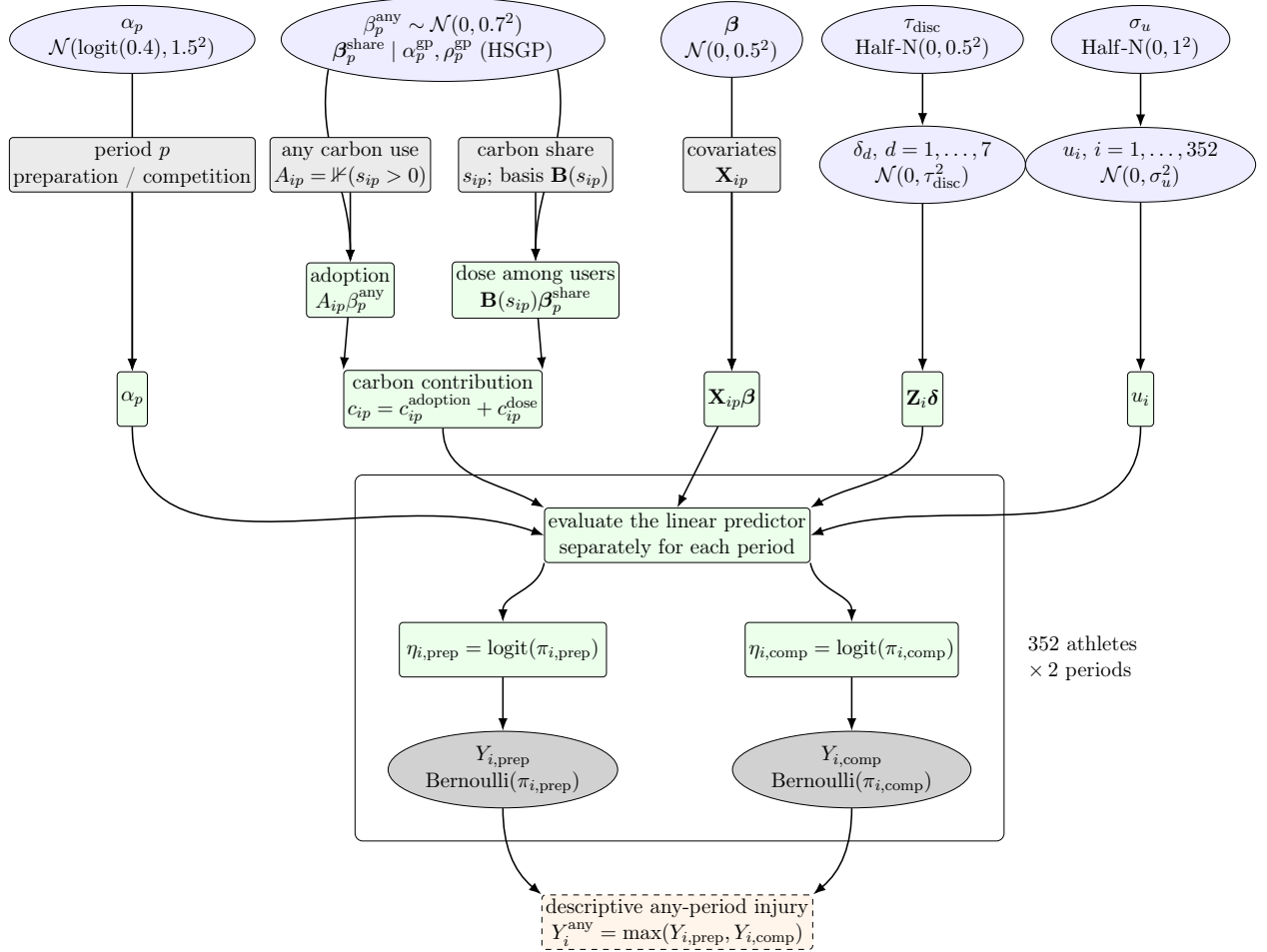


Figure 1: Graphical representation of the primary Bayesian multilevel model. Grey boxes are observed inputs, blue ellipses are estimated parameters, green boxes are deterministic contributions to the linear predictor, and shaded ellipses are the two observed period-specific injury outcomes, jointly enclosed in the solid plate (352 athletes $\times$ 2 periods). The carbon contribution c_{ip} combines adoption (A_{ip} , any carbon use) and dose ($\mathbf{B}(s_{ip})$, the centered Hilbert-space Gaussian-process basis for carbon share among users, with 40 basis functions and period-specific marginal-standard-deviation and length-scale hyperparameters $\alpha_p^{\text{gp}}, \rho_p^{\text{gp}}$ not shown separately for space). $\mathbf{Z}_i$ denotes the seven discipline indicators underlying the partially pooled coefficients δ_d , and u_i the shared athlete random intercept. The common model structure is evaluated separately as $\eta_{i,\text{prep}}$ and $\eta_{i,\text{comp}}$; each predictor points only to its corresponding outcome. The dashed box is the derived any-period injury indicator reported descriptively, outside the probabilistic-model plate, and is not an additional likelihood term.

an injury in either of the two preceding seasons. The period-matched training-frequency scores were fixed at their pooled cohort medians: long endurance 1, medium endurance 1, lactate 2, maximum velocity 2, technique 2, and gym 2; these are the survey’s ordinal frequency scores, not exact counts of unique sessions. The athlete random intercept was fixed at zero (the median of its modeled distribution). Thus, “median profile” denotes medians for continuous and ordinal covariates combined with the modal observed sex and discipline profile, not a uniquely defined median athlete. Adoption was summarized as the risk difference between a non-user and a user at 50% carbon share. Among users, the continuous HSGP risk curve was evaluated over the share range supported in both periods, with preparation and competition predictions calculated for this identical profile.

For display only, each continuous curve was read at three exact share values: 25%, 50%, and 75%. These values are not P25, P50, and P75 of the sample, do not define exposure groups, and are not proposed clinical thresholds. They were selected after model fitting as round, evenly spaced interpretive anchors within the observed support of both periods; 50% is also the pooled median user share and the centering point of the HSGP curve, while 25% and 75% are symmetric around it. A change from 25% to 50%, or from 50% to 75%, therefore means changing the carbon component from one-quarter to one-half, or from one-half to three-quarters, of the same fixed total reported training-frequency score for the same reference profile. Both changes are derived from one continuous fitted function, not from separate models or comparisons between athlete subgroups. These are conditional, profile-specific model predictions intended to aid interpretation of the fitted association, not validated individual prognostic estimates.

2.8.3 Matched comparison of relative and absolute exposure

To examine whether the observed association was specific to relative allocation or instead tracked absolute carbon use, two matched multilevel models represented two substantive exposure hypotheses. The *relative-use hypothesis* used carbon share and asks whether injury prevalence changes as a greater proportion of a given training-frequency total is allocated to carbon-plated spikes. The *absolute-use hypothesis* used the absolute carbon-frequency score and asks whether injury prevalence changes with more reported carbon use after adjustment for the same total training-frequency covariates. Both exposures were standardized among users across both periods (mean 0, standard deviation 1) and entered linearly with period-specific coefficients; each model also retained the period-specific adoption term, identical covariates, discipline effects, priors, and athlete random-intercept structure of the primary model. One standard deviation represented 0.241 in share (24.1 percentage points) or 3.36 units of the summed ordinal carbon-frequency score. The latter is a change in the questionnaire-derived score, not 3.36 unique training sessions, because frequency categories were top-coded and training domains could overlap within a session. The two exposures were fitted in separate models because absolute carbon frequency contributes directly to both the share numerator and the adjusted total-frequency scores; simultaneous inclusion would make the coefficients difficult to identify and interpret.

For a common conditional estimand, posterior injury risk was evaluated from 0.5 standard deviations below to 0.5 standard deviations above the pooled user mean. On the observed scales, this one-standard-deviation increase was from 37.3% to 61.4% carbon share or from 2.83 to 6.19 units of the summed ordinal carbon-frequency score. Continuous covariates were fixed at their cohort medians, discipline at the most prevalent observed indicator, and the athlete random intercept at zero. The comparison was designed to place the two scales on equal footing, not to treat absence of an association in one model as proof of no effect or to select a causal exposure mechanism. This matched exposure-scale comparison was added to clarify interpretation of the primary share analysis and was not part of the prospectively specified sensitivity set.

2.8.4 Validation and sensitivity analyses

Predictive comparison between the primary (exposure) and baseline (no-exposure) models used Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO) [Vehtari et al., 2017], computed on the athlete-grouped joint log-likelihood (both period records per athlete evaluated jointly, with the athlete random intercept marginalized by Gauss–Hermite quadrature before leave-one-athlete-out prediction), reported as the expected log predictive density (ELPD) difference with its standard error. Posterior predictive checks compared observed and model-replicated counts of injured athletes per period. Two pre-specified sensitivity analyses tested structural assumptions of the primary specification: (1) a linear carbon-share term in place of the HSGP curve; (2) wider hyperpriors (HSGP marginal standard deviation $\alpha_p^{\text{gp}} \sim \text{Half-N}(0, 1.2^2)$, covariate coefficients $\mathcal{N}(0, 1.0^2)$, length-scale prior unchanged). Sensitivity models were compared with the primary model as robustness checks on the same estimands, not as competing substantive hypotheses.

2.9 Population reweighting

Because the two-phase recruitment (Section 2.4) could not guarantee equal selection probabilities across age-by-sex strata, a poststratification step [Gelman, 2007] reweighted the analysis sample to FIDAL national population estimates for six age-by-sex strata (Under 20, Under 23, Senior $\times$ Men, Women; Appendix Table 17). Stratum model weights were the ratio of the population share to the observed sample share, further regularized to avoid extreme weights. The Kish effective sample size after weighting was 340.5 of 352 (97%), indicating a modest efficiency loss from reweighting. Weighted estimands were computed by combining the athlete-level posterior predictions of Section 2.8.2 with model weights before marginalization, and are reported alongside the unweighted (sample-representative) estimands as a sensitivity check on generalizability to the national competitive population.

2.10 Subgroup analyses

As a pre-specified descriptive and exploratory extension, the primary adjusted estimands were re-computed conditional on sex (men, women) and on age group (Under 20, Under 23, Senior), holding the same model structure and covariate adjustment, to assess whether the direction and approximate magnitude of the competition-period dose association were consistent across these strata. These subgroup estimates are not powered as confirmatory tests of effect-modification and are reported as descriptive, hypothesis-generating comparisons.

2.11 Sources of bias

The survey instrument underwent content-validity and face-validity review by athletics-medicine content experts and prospective respondents before deployment, but no formal test-retest reliability or criterion-validity assessment against objectively measured training data (for example, coach-logged sessions or shoe-embedded wear sensors) was conducted; this is treated as a limitation rather than a validated measurement property (Section 4.4). Both exposure and outcome were self-reported for the same recall period by the same respondent, which admits recall bias and the possibility that athletes’ beliefs about carbon-plate risk influenced how they reported either quantity. The survey was administered near the end of the competitive season, so the competition-period recall window was recent while the preparation-period window reached further back into the season; this asymmetry is a plausible source of differential, period-specific recall error, and any resulting

attenuation in preparation-period exposure and outcome reporting would bias that period’s association toward the null independently of the underlying biomechanics. This offers one plausible explanation, among others, for why the preparation-period association was weaker and less certain than the competition-period one, and is a reason to treat the period contrast itself, not only the individual period estimates, with some caution. Reporting was not restricted to medically trained staff; athletes reported their own injuries and training without independent clinical verification, following the broad, non-medical-attention-dependent injury definition of Section 2.6, which was chosen specifically to reduce under-ascertainment of injuries that did not prompt a clinical visit at the cost of relying on athlete-reported symptom recognition rather than diagnosis.

2.12 Missing data and other analyses

The analysis used a complete-case cohort by design: eligibility required a fully completed survey, and the three further exclusions in Section 2.4 addressed incomplete injury timing or a structurally undefined carbon-share denominator. No further imputation was performed.

2.13 Software

The fitted models used a Bernoulli likelihood with a logit link. Posterior sampling used Stan’s No-U-Turn Sampler (NUTS), a Hamiltonian Monte Carlo algorithm, through the `rstan` interface. For each model, four independent chains each ran for 2000 iterations; the first 1000 were used for warm-up and adaptation and discarded, leaving 1000 retained draws per chain and 4000 retained draws in total to approximate the posterior distribution. The target acceptance probability was set to `adapt_delta`=0.98 and the maximum tree depth to 12. Runs used base seed 20260719 with deterministic model-specific offsets. Analyses used the `loo` package for PSIS-LOO cross-validation; the HSGP basis, spectral density, and length-scale prior calibration were implemented directly in Stan and base R, with no additional package dependency. Full software versions are reported in Appendix B.

3 Results

3.1 Sample flow and recruitment

Table 3 summarizes the flow from survey completion to the balanced analysis cohort of 352 athletes and 704 athlete-period records. Phase 2 targeted recall concentrated additional enrollment in Under 20 and Under 23 strata, particularly hurdles, horizontal jumps, and combined events, which the voluntary Phase 1 sample under-represented relative to the FIDAL population estimate. After poststratification (Section 2.9; Appendix Table 17), the Kish effective sample size was 340.5 of 352 analyzed athletes (97%), indicating that reweighting did not substantially inflate posterior uncertainty.

3.2 Baseline characteristics

Table 2 compares baseline characteristics by competition sex, the principal descriptive contrast for this cohort, alongside the corresponding population-weighted standardized mean differences (SMD). As expected for a sex comparison, height and weight showed large SMDs (−1.81 and −1.83); demographic, injury-history, carbon-exposure, and discipline-distribution SMDs were mostly below the conventional 0.20 threshold for a small imbalance [Hayes-Larson et al., 2019], with horizontal jumps and combined events showing modest over-representation of women (SMD 0.22–0.23).

Reweighting to national population strata (weighted SMD column) did not materially change this pattern, supporting the sample as broadly representative of the target competitive population once poststratified.

3.3 Descriptive carbon-plate insole use

Across the balanced cohort, carbon-plated spikes accounted for a mean 35.7% of reported preparation-period training-type frequency and 47.7% of reported competition-period training-type frequency (Table 4). This pattern—roughly two-fifths of preparation training and roughly half of competition training reported in carbon-plated spikes—was stable across sex and age-group strata (Table 4, Figure 4), with mean shares differing by at most 4 percentage points between men and women or across the three age groups in either period, and with the full distributions—not only their means—overlapping closely between strata. Adoption (any carbon use) was high in both periods: 81.5% of athletes reported any preparation carbon use and 87.5% reported any competition carbon use (Table 2), consistent with carbon-plated spikes functioning as the prevailing rather than niche equipment choice in this population.

3.4 Specialty-group-stratified characteristics

To summarize the seven disciplines with limited individual cell sizes (19–107 athletes; Appendix Table 14), they were collapsed into four broader, physiologically motivated specialty groups—sprint/hurdles, endurance (middle and long distance), jumps (horizontal jumps or pole vault), and combined events. Tables 5 and 6 report baseline characteristics, injury prevalence, and carbon-plate use by group. Because disciplines and, consequently, groups are multi-select, 26 of 352 athletes belong to more than one group. Two patterns stand out as original descriptive findings of this cohort.

First, competition-period carbon share tracked the explosive-versus-endurance character of the group rather than being uniform across athletics. The sprint/hurdles group (the largest, $n = 169$) had the highest competition carbon share (0.56) and the largest preparation-to-competition rise (+0.17), while the endurance group ($n = 109$) had the lowest carbon share in both periods (0.31 and 0.34). This is consistent with the biomechanical rationale in Section 1: carbon-plate stiffness benefits are most established for short-duration, high-power efforts, and this cohort’s self-reported equipment allocation mirrors that distinction even though the survey did not ask athletes about performance rationale directly.

Second, injury prevalence did not track carbon share in the same way. The jumps group ($n = 62$) stood out with the highest prior-injury history (83.9%) and the highest competition-period injury prevalence (51.6%) of the four groups, despite an intermediate carbon share (0.47)—higher than endurance but lower than sprint/hurdles—indicating that the elevated injury burden in jumping disciplines is unlikely to be explained by carbon-plate dose alone and more plausibly reflects the landing and eccentric loading demands specific to jumping events; the primary Bayesian model (Section 3.7) already adjusts for discipline using partially pooled indicators specifically to avoid attributing this discipline-driven variation to carbon exposure. The combined-events group ($n = 41$) showed a distinct hybrid profile: a carbon share close to sprint/hurdles (0.52) paired with an endurance-like training volume (preparation and competition medians of 9) and an elevated any-injury prevalence (70.7%), consistent with the mixed physiological demands of multi-event competition and with the higher, more heterogeneous musculoskeletal load noted for plyometric and multi-skill disciplines in prior athletics injury literature [Edouard et al., 2020].

3.5 Injury location and type

The 204 injured athletes reported 259 injuries in total (149 athletes with one injury, 55 with two), a median of 1 injury per affected athlete. Table 7 reports the anatomical location of these injuries and Figure 2 displays both the location and type distributions together. The thigh was, by a substantial margin, the single most commonly reported location (97 of 259 injuries, 37.5%; affecting 80 of 204 injured athletes, 39.2%), followed by the lower leg (25.1% of injuries) and the foot (19.3%); together, the hip, groin, thigh, lower leg, ankle, and foot accounted for 95.8% of all reported injuries, with only 4.2% localized to the head, upper limb, or trunk.

For the 248 lower-limb injuries for which an injury type was recorded (Table 8, Figure 2), tendinopathy (28.6%) and bone stress injury or stress fracture (26.2%) were the two most frequently reported types, together accounting for more than half of classified lower-limb injuries and each individually more common than muscle strain (22.2%). This is a distinctive feature of this cohort relative to general athletics injury surveillance, where muscle injury is typically the most frequently reported tissue type by a wide margin and tendon and bone injuries are reported considerably less often [Edouard et al., 2020].

Cross-tabulating type against location (Table 9) shows that these two leading types were not distributed the same way anatomically. Tendinopathy was disproportionately distal relative to the overall injury-location pattern: foot (35.2% of the 71 tendinopathy injuries) and lower leg (33.8%) were more prominent locations for tendinopathy than for the injury sample as a whole (19.3% and 25.1% respectively; Table 7), and ankle involvement (22.5%) was likewise higher than the sample-wide 13.9%—a pattern compatible with Achilles- and plantar-tendon structures at the distal lower limb, consistent with the biomechanical literature on carbon-plate loading of the ankle and forefoot [Gerlach et al., 2025]. Bone stress injury or stress fracture, in contrast, was thigh-dominant (58.5% of the 65 bone stress injuries), a pattern closely tracking the overall injury-location distribution and similar to muscle strain (74.5% thigh) rather than being disproportionately distal; foot involvement among bone stress injuries (15.4%) was, if anything, lower than the sample-wide rate. The elevated overall frequency of bone stress injury in this cohort is therefore a genuine descriptive finding, but this cohort’s data do not show the foot/navicular-specific concentration that would most directly support the mechanistic link to carbon-plate loading proposed for that injury type [Tenforde et al., 2023]; the survey’s type item does not further subclassify bone site (for example, navicular versus tibial versus metatarsal), so this distinction cannot be resolved further with the present data. Taken together, the location-by-type pattern strengthens the tendinopathy–carbon-plate hypothesis specifically, while leaving the bone-stress-injury hypothesis only partially supported. This is reported as a hypothesis-generating observation: the present cross-sectional design and sample size do not support a formal type-stratified test of the carbon-share association, and confirming a tendinopathy-specific association with carbon-plate dose, or resolving the anatomical site of the bone stress injuries observed here, would require a dedicated, adequately powered follow-up.

3.6 Athlete-reported injury burden

Of the 204 injured athletes, 193 (94.6%) completed all four adapted OSTRC-H2 impact items. The median total severity score was 75 (IQR 58–92) overall (Table 1). Scores were similar for athletes injured only during preparation (median 71.5, IQR 50–90) and only during competition (67, IQR 58–84; SMD -0.07), while athletes reporting injury in both periods had a higher median score of 84 (IQR 67–92; SMD 0.41 versus competition only). The largest component difference was for symptoms in the both-period group (SMD 0.51). These results describe impact among athletes already classified as injured; they are not a second injury-prevalence analysis.

Table 1: Adapted OSTRC-H2 impact scores among 193 injured athletes with complete responses, stratified by mutually exclusive injury period. Values are median [Q1, Q3]. Each component ranges from 0 to 25 and total severity from 0 to 100; higher scores indicate greater impact. SMDs compare group means with the competition-only group and are descriptive.

Domain	Overall (<i>n</i> = 193)	Preparation only (<i>n</i> = 50)	Competition only (<i>n</i> = 84)	Both periods (<i>n</i> = 59)	SMD vs competition	
					Preparation	Both
Participation	17 [17, 25]	17 [17, 25]	17 [8, 25]	17 [17, 25]	0.14	0.19
Training volume	17 [17, 25]	17 [8, 25]	17 [8, 25]	25 [17, 25]	0.01	0.29
Performance	17 [17, 25]	17 [8, 25]	17 [17, 25]	25 [17, 25]	−0.35	0.22
Symptoms	17 [8, 25]	17 [8, 17]	17 [8, 17]	17 [17, 25]	−0.01	0.51
Total severity	75 [58, 92]	71.5 [50, 90]	67 [58, 84]	84 [67, 92]	−0.07	0.41

Injury location and type in the balanced cohort

Multi-select items; percentages sum to more than 100% within each panel

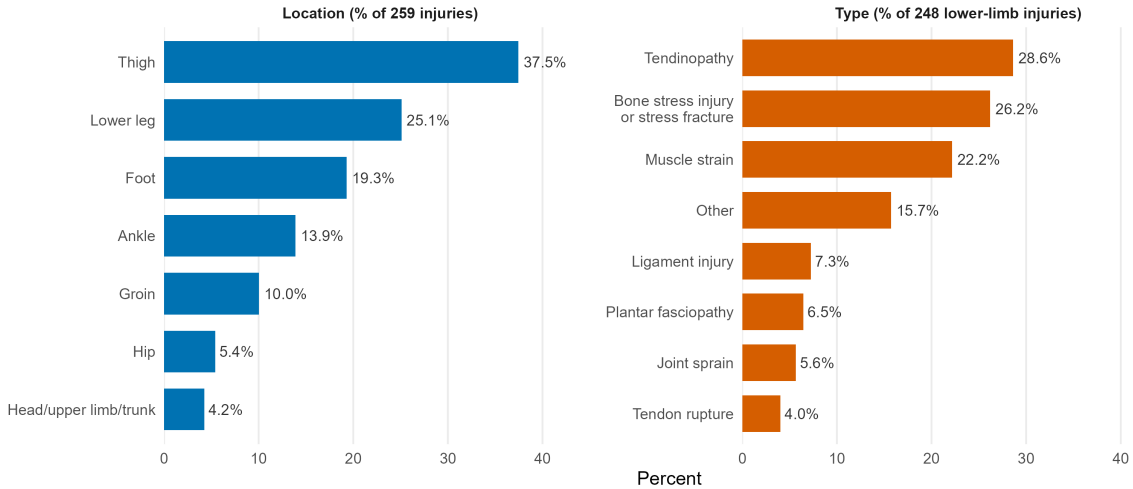


Figure 2: Anatomical location (% of 259 injuries) and type (% of 248 lower-limb injuries with a recorded type) of injuries reported by the balanced analysis cohort. Both items are multi-select, so percentages within each panel sum to more than 100%.

3.7 Primary adjusted associations

Dose is represented by one continuously varying curve per period, estimated with a Hilbert-space Gaussian process (HSGP, Matérn 5/2; Section 2.8.1). The estimates in this section are population-standardized, averaged over the observed covariate and discipline distribution of the full period-specific sample and over the estimated athlete-to-athlete heterogeneity: they describe injury prevalence for this population under each exposure scenario. Figure 3 shows the observed distribution of carbon share by period, Figure 6 compares adoption with non-adoption in a common reference profile, Figure 7 shows the fitted curve itself—the population-standardized version in its top row, individual posterior draws of that same curve layered underneath it—and Table 10 reports the primary estimands, pre-specified point summaries of that curve read at period-specific empirical percentiles rather than fixed 25% and 75% share values.

In preparation, the curve is essentially flat across the observed share range once adjusted for covariates: adjusted injury prevalence was 29.5% (95% CrI 20.5–39.5%) for non-users and 36.1% (95% CrI 29.1–44.9%) for the P50 user (42.1% share), an adoption difference of 6.5 percentage points (pp) (95% CrI –5.6 to 19.2; $P_{\text{dir}} = 85.3\%$); the P75-versus-P25 dose contrast (57.1% vs. 28.6% share) was 0.0 pp (95% CrI –12.0 to 10.2; $P_{\text{dir}} = 50.6\%$). Neither preparation-period contrast reached the directional certainty used here to describe an association as clear.

Competition is where the curve does real work. Adjusted competition-period injury prevalence was 37.5% (95% CrI 25.9–49.8%) for non-users and rose to 42.3% (95% CrI 34.3–49.7%) at P50 share (50.0%), an adoption difference of 4.7 pp (95% CrI –9.6 to 18.6; $P_{\text{dir}} = 74.7\%$). Among users, moving continuously from P25 (36.0% share) to P75 (71.4% share) tracked a rise in adjusted prevalence from 37.6% (95% CrI 28.2–46.2%) to 50.8% (95% CrI 41.1–62.0%): a difference of 13.2 pp (95% CrI –0.2 to 29.1; $P_{\text{dir}} = 96.9\%$) and a ratio of 1.34 (95% CrI 1.00 to 1.99; $P_{\text{dir}} = 96.9\%$), with an 83.6% posterior probability of exceeding the pre-specified 5-pp practical-relevance threshold. This is the strongest of the four primary population-level contrasts by a wide margin: it is the only one clearing the practical-relevance threshold, and the continuous curve shows it building steadily with share across the full observed range. The individual posterior draws in the top-right panel of Figure 7 make the same point visually: nearly all of them climb from low to high share, even though they disagree on how steeply, which is a different and stronger kind of consistency than the pointwise credible band alone can show—compare the top-left (preparation) panel, where the draws fan out in every direction with no shared trend. Its 95% interval marginally includes zero, and Section 3.10 shows the same pattern under two structurally different specifications, so we read this as a real, population-level, clinically actionable signal whose precise size current data cannot yet pin down.

Figure 6 and the fixed-profile panel of Figure 7 also read the fitted curves for one reference profile (a male speed athlete aged 21 years, 176 cm tall and weighing 64 kg, with 10 years of athletics experience and a prior injury, training-frequency scores at their pooled medians, and athlete random effect zero), turning the population-averaged association above into concrete absolute-risk numbers for one illustrative athlete; this single-profile reading is secondary and descriptive, and the primary conclusions of this section rest on the population-standardized estimates above. For that profile, competition-period risk at 25%, 50%, and 75% share was 46.5%, 54.3%, and 64.3% respectively, again a steady climb (75%-minus-25% difference 17.7 pp, 95% CrI –0.3 to 37.1, $P_{\text{dir}} = 96.9\%$), while the same readings in preparation were flat at 45.0%, 47.5%, and 45.0%. Here, as throughout, 25%, 50%, and 75% are exact share values chosen as readable anchors on the continuously estimated curve. The matched comparison below tests whether the broader competition pattern is specific to relative share.

3.8 Relative versus absolute carbon exposure

The matched linear models did not support interpreting the competition association as specific to carbon share (Figure 8). Among competition-period users, a one-standard-deviation increase in carbon share—24.1 percentage points, evaluated from 37.3% to 61.4%—was associated with a 7.9-pp higher profile-specific injury risk (95% CrI 1.4 to 14.4; $P_{\text{dir}} = 99.0\%$). A one-standard-deviation increase in absolute carbon-frequency score—3.36 summed ordinal score units, evaluated from 2.83 to 6.19—with the identical adjustment for period-matched total training frequency, was associated with a similar 8.3-pp increase (95% CrI 1.6 to 15.9; $P_{\text{dir}} = 99.2\%$). The score increment does not mean 3.36 unique sessions. The corresponding conditional odds ratios were 1.38 (95% CrI 1.06 to 1.80; $P_{\text{dir}} = 99.0\%$) and 1.41 (95% CrI 1.07 to 1.92; $P_{\text{dir}} = 99.2\%$), respectively.

Neither exposure scale showed a clear preparation-period association: for a share increase of 24.1 percentage points, the risk difference was 3.6 pp (95% CrI -3.9 to 11.2 ; $P_{\text{dir}} = 82.8\%$), while for an absolute-score increase of 3.36 units it was 1.5 pp (95% CrI -6.9 to 9.4 ; $P_{\text{dir}} = 63.2\%$). Thus, the comparable positive competition estimates support a general higher-carbon-exposure interpretation, but do not distinguish a relative-allocation mechanism from an absolute-frequency mechanism. The standardized linear contrasts are not numerically interchangeable with the primary HSGP model’s 75th-versus-25th-percentile contrast: they were constructed specifically to compare the two exposure scales under the same functional form and observed-scale increments defined above.

3.9 Model validation

Both the primary (exposure) and matched baseline (no-exposure) HSGP-era models showed acceptable sampling behavior: maximum $\hat{R} \leq 1.008$, zero divergent transitions, zero maximum-tree-depth hits, and minimum E-BFMI of 0.60 across both models (Appendix Table 20). Posterior predictive checks reproduced the observed injury counts closely: the full posterior predictive distribution of replicated injured-athlete counts was centered on 115.6 (observed 115) for preparation and 149.7 (observed 149) for competition, with the observed count falling centrally within each replicated distribution (Figure 5). Every athlete-level PSIS-LOO Pareto- k value was at most 0.5 in both models, supporting the reliability of the leave-one-out estimates. The exposure model’s athlete-grouped ELPD exceeded the baseline model’s by 1.92 (SE 2.41) (Appendix Table 20): the exposure terms did not clearly improve held-out predictive performance, even though the competition-period dose contrast itself had a comparatively high posterior probability of being positive. These two findings are compatible: a single, period-specific exposure effect concentrated in a subset of the competition-period users can produce a suggestive marginal association without moving whole-model out-of-sample predictive accuracy by a detectable amount in a cohort of this size.

The two matched exposure-scale models also sampled adequately: maximum $\hat{R}$ was 1.007, minimum effective sample size was 531, and neither model produced a divergent transition or maximum-tree-depth hit.

3.10 Sensitivity analyses

The direction of the competition-period dose association was stable across both pre-specified sensitivity analyses, and so was its credible interval sitting just at the zero boundary (Table 11, Figure 9). Under a linear carbon-share parameterization, the association reduced to 5.9 pp (95% CrI -0.3 to 12.6 ; $P_{\text{dir}} = 96.9\%$). Under wider HSGP hyperpriors, the association increased to 14.6 pp (95% CrI -0.1 to 30.7 ; $P_{\text{dir}} = 96.9\%$), close to the primary HSGP estimate of 13.2 pp (95% CrI -0.2 to 29.1 ; $P_{\text{dir}} = 96.9\%$). All three specifications place at least 96.9% of their posterior mass on a

positive association, converging on a plausible magnitude somewhere in the 6–15 pp range. This consistency across three structurally different specifications supports treating the competition-period dose association as a genuine, data-supported feature of these data. The HSGP estimates how much curvature the dose–response relationship needs directly from the data, and the resulting interval width is an honest reflection of that estimation uncertainty, giving a directionally reliable signal (about 97% probability) whose precise magnitude is still being resolved.

3.11 Population-weighted and subgroup estimates

After poststratification to national age–sex population strata, the competition-period dose association from the primary HSGP model changed little: 13.7 pp weighted (95% CrI -0.2 to 30.0 ; $P_{\text{dir}} = 96.8\%$) versus 13.2 pp unweighted (95% CrI -0.2 to 29.1 ; $P_{\text{dir}} = 96.9\%$) (Appendix Table 19, Figure 10, Figure 11). The two adoption contrasts and the preparation-period dose contrast likewise remained centered near no association after weighting. Conditional on sex or age group, the competition-period dose association retained a consistent point estimate (13.1–14.1 pp) and an essentially unchanged P_{dir} of 96.8% in every stratum examined (Appendix Table 18), providing no descriptive evidence that the association was concentrated in, or specific to, a single sex or age-group stratum.

Observed carbon-frequency share

All 352 athletes; zero marks nonuse and one means all reported type-frequency was carbon

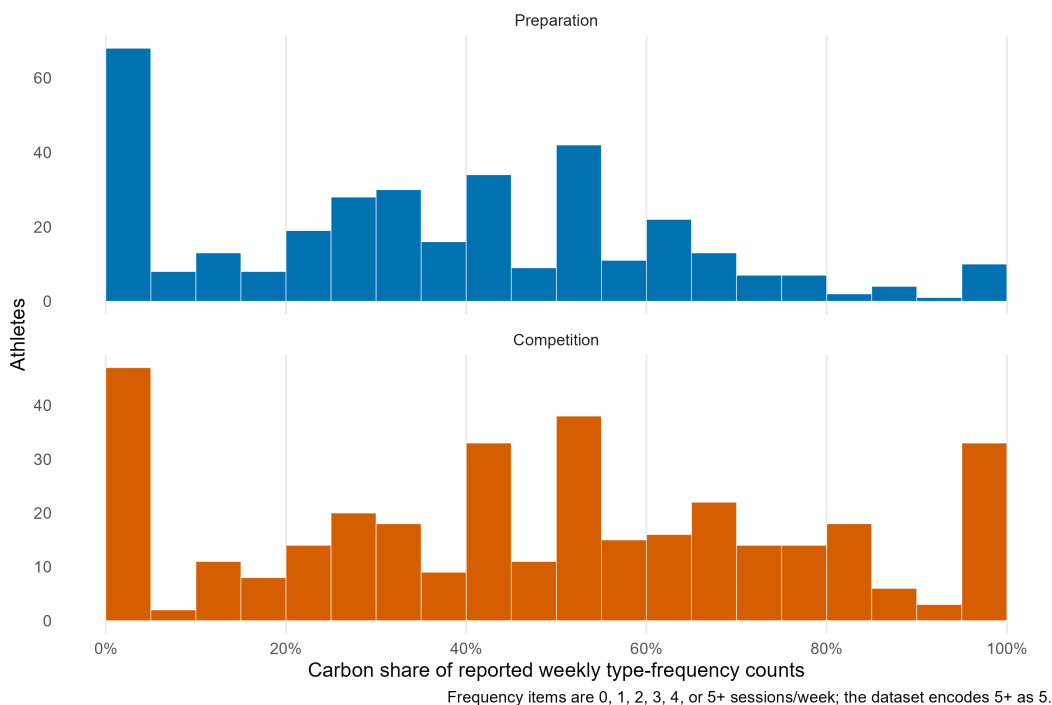


Figure 3: Observed distribution of carbon-plate insole share of reported training-type frequency, among all 352 athletes, by period.

Carbon-plate share of training is stable across sex and age group

Observed carbon share of reported training-type frequency, by period, age group, and sex



Figure 4: Observed distribution of carbon-plate insole share of reported training-type frequency, by period, age group, and sex. Teal denotes men and pink denotes women; preparation and competition are separated by rows rather than encoded by colour. Curves are boundary-corrected kernel density estimates over the observed discrete-ratio share values (Section 2.5), shown to compare distributional overlap across strata; they are a smoothing device for visualization.

Posterior predictive distribution of injured-athlete counts

Replicated counts from the primary HSGP model (4000 posterior draws) versus the observed count, by period

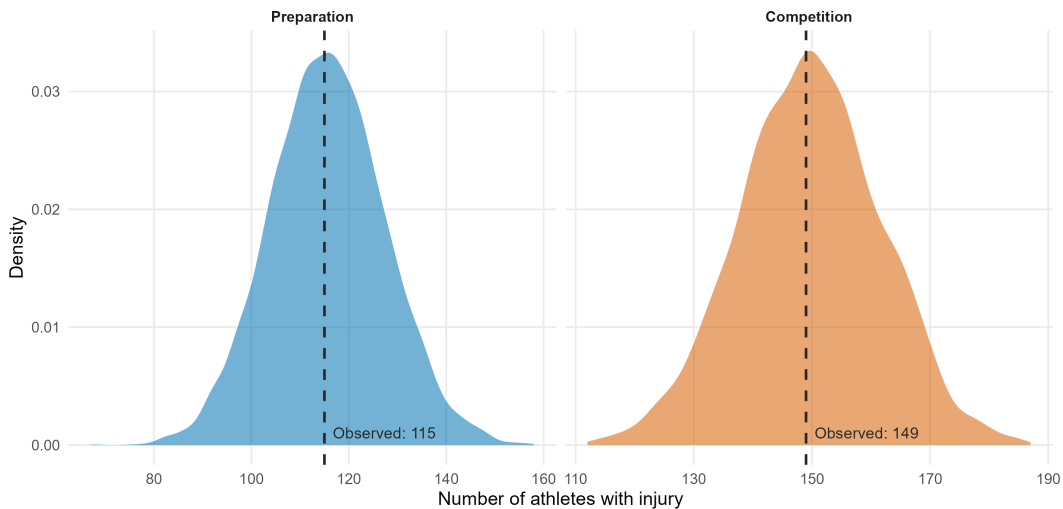


Figure 5: Posterior predictive check: full posterior distribution of the model-replicated number of injured athletes (4000 posterior draws of the primary model), by period. Dashed vertical line: observed count.

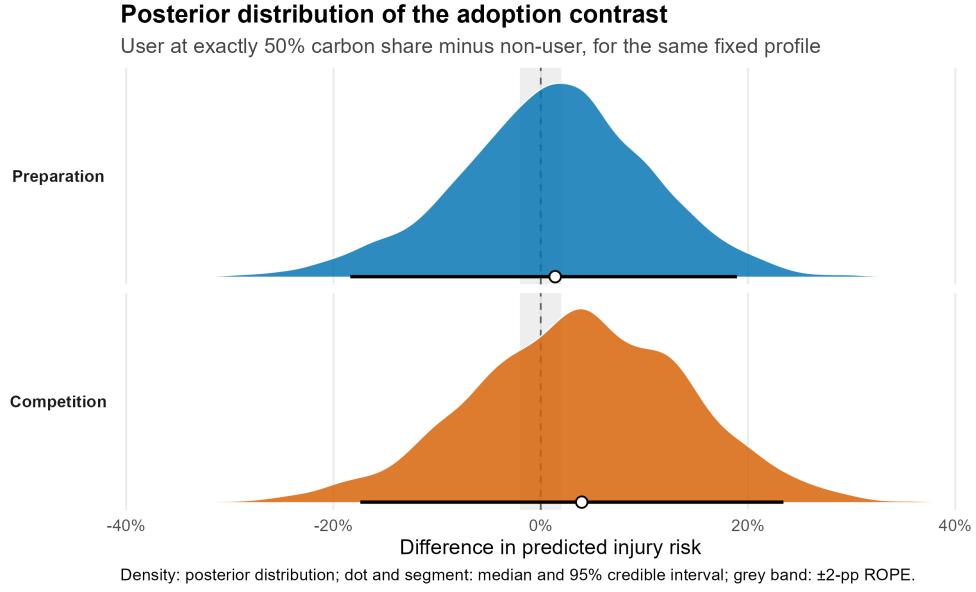


Figure 6: Posterior distributions of the profile-specific adoption contrast (user at exactly 50% carbon share minus non-user), by period. Filled curves show posterior densities; points and horizontal segments show posterior medians and 95% credible intervals; the grey band marks the ± 2 -percentage-point region of practical equivalence and the dashed line marks zero. The fixed reference profile is a male speed athlete aged 21 years, 176 cm tall, weighing 64 kg, with 10 years of athletics experience, a prior injury, training-frequency scores at their pooled medians, and athlete random effect zero; corresponding P_{dir} values are reported in the text.

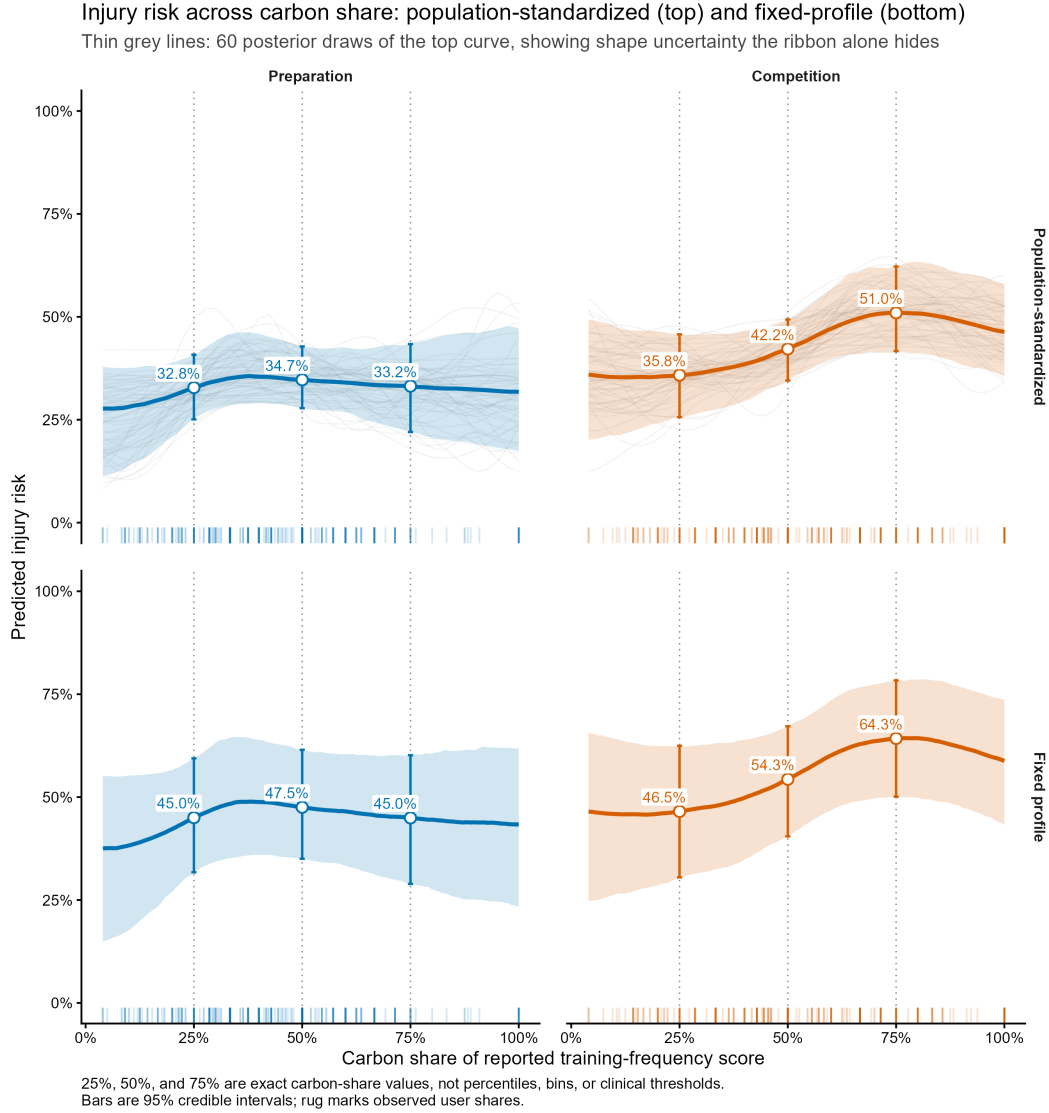


Figure 7: Fitted injury risk across carbon share among users, from the Hilbert-space Gaussian-process (HSGP, Matérn 5/2) curve. Top row: population-standardized estimand, averaged over the observed covariate and discipline distribution of the full sample (the primary estimand of Section 3.7); thin grey lines are 60 individual posterior draws of this same curve, illustrating how much the curve’s *shape*—not just its level at any one point—remains uncertain, information the credible band alone does not convey. Bottom row: the identical curve read for one fixed male speed-athlete reference profile, a secondary, descriptive illustration (Figure 6 caption). In both rows, the marked 25%, 50%, and 75% values are exact shares selected as round, evenly spaced interpretive anchors within both periods’ observed support; they are not empirical percentiles, bins, fitted knots, clinical thresholds, or separate subgroup comparisons. Points and labels show posterior median risk at each anchor, vertical bars and coloured bands show 95% credible intervals, and rugs show observed user shares. The formal primary percentile contrast instead uses period-specific P25 and P75 values: 28.6% and 57.1% in preparation, and 36.0% and 71.4% in competition; corresponding P_{dir} values are reported in the text.

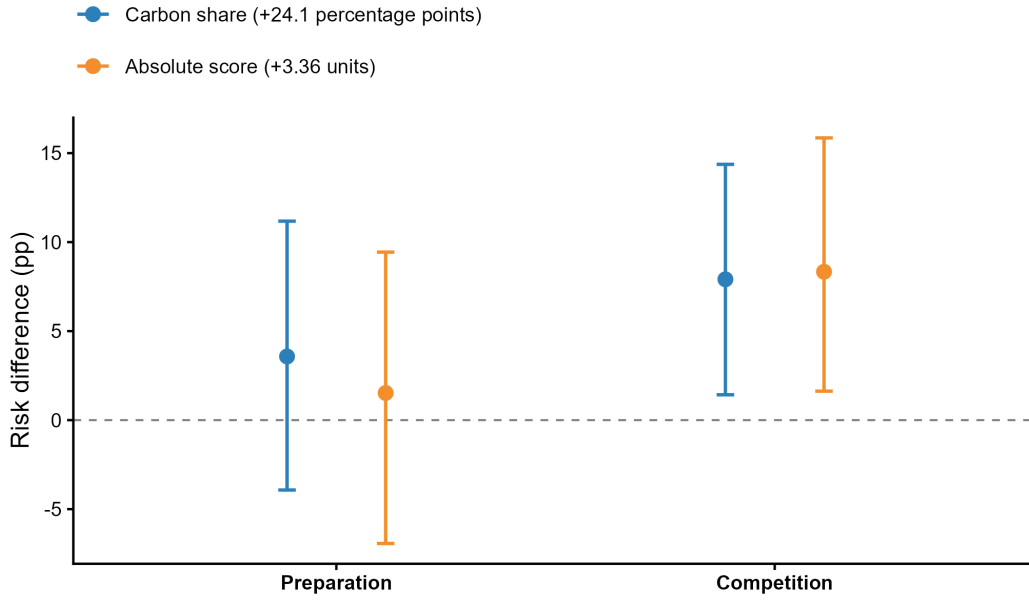


Figure 8: Matched comparison of relative and absolute carbon exposure among users. Points are posterior median profile-specific injury-risk differences for a one-standard-deviation increase; bars are 95% credible intervals. On the observed scales, the contrast is 37.3% to 61.4% carbon share (a 24.1-percentage-point increase) or 2.83 to 6.19 summed ordinal carbon-frequency-score units (a 3.36-unit increase, not 3.36 unique sessions). Both models use the same linear functional form, period-specific exposure coefficients, adjustment covariates, discipline effects, and athlete random-intercept structure; predictions use the same fixed male speed-athlete profile described in Figure 6. Corresponding P_{dir} values are reported in the text.

Full posterior distribution of the competition association

Q75 vs Q25 carbon share among competition-period users

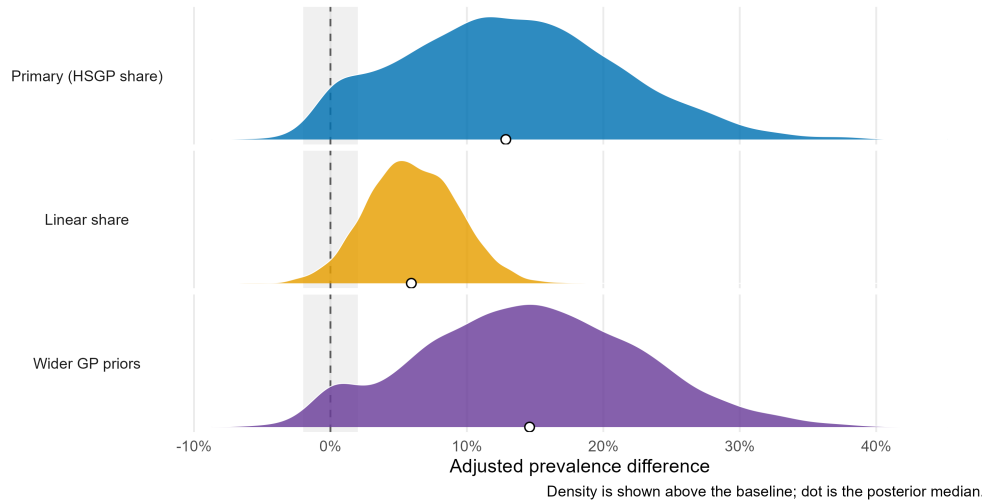


Figure 9: Sensitivity analysis: full posterior distribution of the competition-period empirical P75-versus-P25 dose association (71.4% vs. 36.0% carbon share among competition users) under the primary HSGP specification, a linear share specification, and wider HSGP hyperpriors. This is distinct from the exact 75%-versus-25% interpretive anchors in Figure 7.

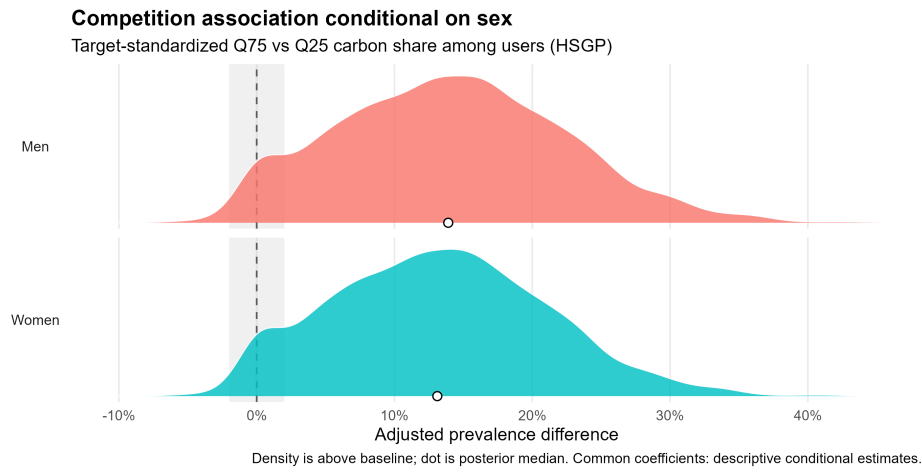


Figure 10: Competition-period dose association, population-weighted and stratified by sex.

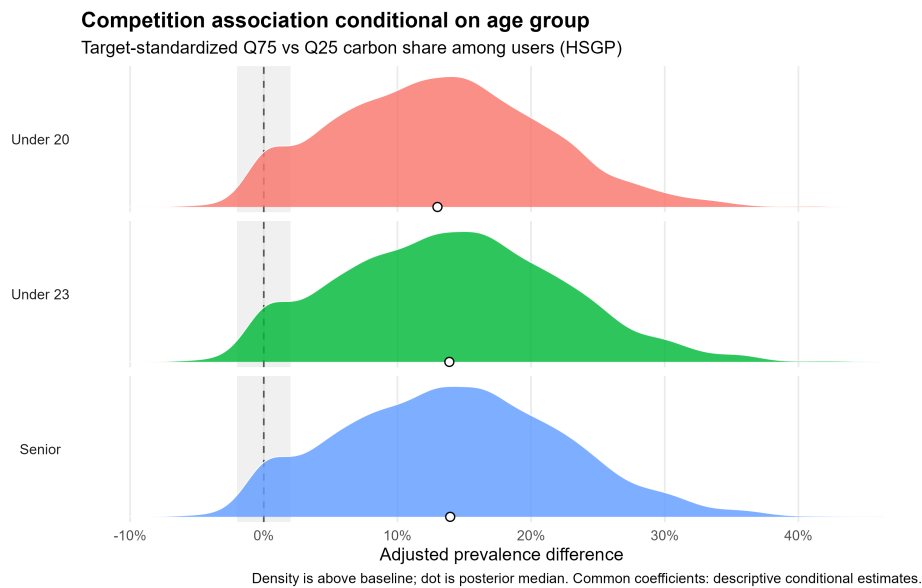


Figure 11: Competition-period dose association, population-weighted and stratified by age group.

Table 2: Baseline characteristics of the balanced analysis cohort ($N = 352$), by competition sex. Values are mean (SD), median [Q1, Q3], or n (%). SMD is the standardized mean difference, women vs. men (Cohen’s d for continuous/ordinal rows, computed on log-odds for proportions); $|\text{SMD}| \geq 0.20$ is conventionally a small imbalance and ≥ 0.50 a moderate one [Hayes-Larson et al., 2019]. Population-weighted columns apply the age-sex poststratification weights of Section 2.9 (Appendix Table 17).

Characteristic	Statistic	Men ($n = 194$)	Women ($n = 158$)	SMD sample	SMD weighted
Demographics					
Age, years	Mean (SD)	21.3 (3.1)	21.9 (3.1)	0.198	0.137
Age group: Under 20	n (%)	71 (36.6)	43 (27.2)	−0.202	−0.144
Age group: Under 23	n (%)	63 (32.5)	53 (33.5)	0.023	0.060
Age group: Senior	n (%)	60 (30.9)	62 (39.2)	0.175	0.093
Anthropometrics					
Height, cm	Mean (SD)	180.6 (5.9)	169.2 (6.8)	−1.809	−1.749
Weight, kg	Mean (SD)	70.5 (7.8)	57.8 (6.0)	−1.831	−1.847
Athletics history					
Experience, years	Mean (SD)	10.1 (4.4)	11.0 (3.9)	0.225	0.220
Injury					
Prior injury (previous 2 seasons)	n (%)	156 (80.4)	111 (70.3)	−0.237	−0.250
Any current-season injury	n (%)	120 (61.9)	84 (53.2)	−0.177	−0.210
Preparation-period injury	n (%)	69 (35.6)	46 (29.1)	−0.138	−0.165
Competition-period injury	n (%)	88 (45.4)	61 (38.6)	−0.137	−0.139
Carbon-plate insole exposure					
Any preparation carbon use	n (%)	159 (82.0)	128 (81.0)	−0.024	−0.041
Any competition carbon use	n (%)	173 (89.2)	135 (85.4)	−0.112	−0.130
Preparation carbon share (among users)	Median [Q1,Q3]	0.4 [0.2, 0.6]	0.4 [0.3, 0.6]	0.125	0.115
Competition carbon share (among users)	Median [Q1,Q3]	0.5 [0.3, 0.7]	0.5 [0.4, 0.8]	0.184	0.161
Training frequency (period total)					
Preparation total type-frequency	Median [Q1,Q3]	8 [6, 11]	8 [6, 13]	0.098	0.069
Competition total type-frequency	Median [Q1,Q3]	7 [5, 10]	7 [5, 11]	0.121	0.106
Discipline					
Sprint	n (%)	64 (33.0)	43 (27.2)	−0.126	−0.122
Middle distance	n (%)	38 (19.6)	31 (19.6)	0.001	0.015
Long distance	n (%)	36 (18.6)	23 (14.6)	−0.108	−0.064
Hurdles	n (%)	51 (26.3)	45 (28.5)	0.049	0.023
Horizontal jumps	n (%)	18 (9.3)	26 (16.5)	0.216	0.183
Pole vault	n (%)	10 (5.2)	9 (5.7)	0.024	0.001
Combined events	n (%)	16 (8.2)	25 (15.8)	0.234	0.195

Table 3: Sample flow from survey respondents to the balanced analysis cohort.

Stage	n
Consented, eligible, complete survey	355
Excluded: incomplete injury timing	−1
Excluded: zero total eligible frequency in a period	−2
Balanced analysis cohort (athletes)	352
Athlete-period records (preparation + competition)	704
Preparation-period injuries	115
Competition-period injuries	149

Table 4: Mean reported carbon share of training-type frequency, overall and by sex and age group. Carbon share is defined in Section 2.5 and includes non-users (share = 0).

Stratum	Level	n	Preparation, mean share	Competition, mean share
Overall	Overall	352	0.357	0.477
Sex	Men	194	0.349	0.468
	Women	158	0.366	0.488
Age group	Under 20	114	0.365	0.453
	Under 23	116	0.379	0.489
	Senior	122	0.328	0.488

Table 5: Characteristics and injury prevalence by specialty group. Groups collapse the seven athletics disciplines (Appendix Table 14) into four physiologically motivated categories: Sprint/hurdles (sprint or hurdles), Endurance (middle or long distance), Jumps (horizontal jumps or pole vault), and Combined events. Because disciplines and, consequently, groups are multi-select, 26 of 352 athletes belong to more than one group.

	<i>n</i>	Age, y	Women, %	Height, cm	Weight, kg	Exp., y	Prior inj., %	Any inj., %	Comp. inj., %
Sprint/hurdles	169	21.4 (3.0)	42.6	175.8 (8.7)	66.9 (9.3)	10.0 (3.9)	75.7	60.9	44.4
Endurance	109	21.6 (3.2)	42.2	174.0 (8.2)	59.6 (7.9)	10.3 (4.3)	67.9	44.0	30.3
Jumps	62	22.2 (3.5)	54.8	175.6 (9.0)	65.8 (9.6)	11.5 (4.6)	83.9	64.5	51.6
Combined events	41	22.0 (3.0)	61.0	176.6 (7.4)	67.8 (8.3)	11.9 (4.4)	82.9	70.7	48.8

Table 6: Carbon-plate insole use and training-type frequency by specialty group (grouping as in Table 5).

Specialty group	Prep. carbon share	Comp. carbon share	Prep. total freq., Mdn	Comp. total freq., Mdn
Sprint/hurdles	0.39	0.56	8	7
Endurance	0.31	0.34	10	9
Jumps	0.33	0.47	6	5
Combined events	0.41	0.52	9	9

Table 7: Anatomical location of reported injuries (259 injuries among 204 injured athletes; multi-select item, so location counts sum to more than 259).

Location	Injuries, n	% of injuries	% of injured athletes ($n = 204$)
Thigh (anterior or posterior)	97	37.5	39.2
Lower leg (anterior or posterior)	65	25.1	27.9
Foot (anterior or posterior)	50	19.3	22.1
Ankle (medial or lateral)	36	13.9	15.2
Groin	26	10.0	12.3
Hip	14	5.4	6.9
Head, upper limb, or trunk	11	4.2	5.4

Table 8: Type of reported lower-limb injuries. The type item was asked only for injuries localized to the hip, groin, thigh, lower leg, ankle, or foot (248 of 259 injuries; the questionnaire ended after the location item for the 11 head/upper-limb/trunk injuries); multi-select item, so counts sum to more than 248.

Type	Injuries, n	% of lower-limb injuries	% of injured athletes ($n = 204$)
Tendinopathy	71	28.6	29.9
Bone stress injury or stress fracture	65	26.2	25.0
Muscle strain	55	22.2	26.5
Other	39	15.7	16.7
Ligament injury	18	7.3	8.8
Plantar fasciopathy	16	6.5	7.8
Joint sprain	14	5.6	6.9
Tendon rupture	10	4.0	3.9

Table 9: Anatomical location of the three leading injury types (% of injuries of that type; multi-select, so rows sum to more than 100%). Location categories with 0% in all three rows (hip, groin, head/upper limb/trunk) are omitted.

Type (n injuries)	Thigh	Lower leg	Ankle	Foot
Tendinopathy (71)	22.5	33.8	22.5	35.2
Bone stress injury/stress fracture (65)	58.5	26.2	7.7	15.4
Muscle strain (55)	74.5	27.3	3.6	0.0

Table 10: Primary adjusted associations between carbon-plate insole exposure and injury prevalence, by period and contrast, from the primary Hilbert-space Gaussian-process (HSGP) share model. Adjusted prevalence difference and ratio are posterior-standardized (Section 2.8.2); credible intervals are 95% equal-tailed and P_{dir} is the posterior probability of the median estimate’s direction. Adoption compares the empirical P50 user with a non-user; dose compares empirical P75 with P25 among users. The percentile shares are 28.6%, 42.1%, and 57.1% in preparation and 36.0%, 50.0%, and 71.4% in competition; they are distinct from the exact 25%, 50%, and 75% display anchors in Figure 7.

Period	Contrast	Difference, pp (95% CrI)	Ratio (95% CrI)	P_{dir}
Preparation	Adoption	6.4 (−5.6 to 19.2)	1.22 (0.85 to 1.85)	85.3%
Preparation	Dose	0.0 (−12.0 to 10.2)	1.00 (0.70 to 1.35)	50.6%
Competition	Adoption	4.7 (−9.6 to 18.6)	1.13 (0.80 to 1.70)	74.7%
Competition	Dose	13.2 (−0.2 to 29.1)	1.34 (1.00 to 1.99)	96.9%

Table 11: Sensitivity of the competition-period dose association (75th vs. 25th percentile share among users) to model specification. P_{dir} is the posterior probability of the median estimate’s direction.

Model	Difference, pp (95% CrI)	P_{dir}
Primary (HSGP share)	13.2 (−0.2 to 29.1)	96.9%
Linear share	5.9 (−0.3 to 12.6)	96.9%
Wider HSGP priors	14.6 (−0.1 to 30.7)	96.9%

4 Discussion

4.1 Key results

This national, two-phase-recruited sample of competitive Italian athletics athletes shows carbon-plated spikes established as the prevailing training equipment: a mean 35.7% of preparation-period and 47.7% of competition-period training-type frequency, with adoption above 80% in both periods and a consistent pattern across sex and age-group strata. Adoption was not clearly associated with injury prevalence in either period. Among competition-period users, the primary nonlinear share model showed higher adjusted injury prevalence at higher share, but the matched linear comparison showed that this association was not unique to the relative scale: an increase from 37.3% to 61.4% share (24.1 percentage points) corresponded to a 7.9-pp risk difference (95% CrI 1.4 to 14.4; $P_{\text{dir}} = 99.0\%$), while an increase from 2.83 to 6.19 units of the summed ordinal carbon-frequency score (3.36 units, not unique sessions) corresponded to 8.3 pp (95% CrI 1.6 to 15.9; $P_{\text{dir}} = 99.2\%$). Both preparation estimates remained uncertain. The adapted OSTRC-H2 items added an athlete-centered burden estimate beyond injury counts: median severity was 75 of 100 and was highest among athletes reporting injuries in both periods. A further original descriptive finding reinforces this pattern: among the 248 lower-limb injuries with a recorded type, tendinopathy (28.6%) and bone stress injury or stress fracture (26.2%) were the two leading injury types, together exceeding muscle strain (22.2%)—a distribution markedly different from general athletics injury surveillance, where muscle injury typically predominates [Edouard et al., 2020]. Cross-tabulating type against location refines this further: tendinopathy was disproportionately concentrated at the foot, lower leg, and ankle relative to the injury sample as a whole, a distal pattern that lines up with the Achilles- and forefoot-tendon loading flagged by the biomechanical literature on carbon-plate use (Section 4.3); bone stress injury, by contrast, was thigh-dominant like the sample overall, so its elevated frequency in this cohort is a genuine finding but is not, on the present data, specifically localized to the foot or ankle.

4.2 Interpretation

The distinction between adoption and amount of use remains central: the association tracked greater competition-period exposure among users, not simply adoption. The exposure-scale comparison, however, changes the mechanistic interpretation. Similar standardized estimates for share and absolute frequency mean that these data cannot identify whether relative allocation to carbon footwear or cumulative carbon-use frequency is the more relevant quantity. The two measures are mathematically related and were therefore fitted separately; adjustment for period-matched total training frequency makes each coefficient conditional on reported training volume but does not turn their descriptive comparison into causal separation. The result is compatible with a cumulative-load mechanism in which greater exposure to altered longitudinal bending stiffness changes ankle and plantarflexor loading [Gerlach et al., 2025, Willwacher et al., 2016], but it is also compatible with unmeasured competition-period intensity, footwear allocation, or reverse causation.

The primary HSGP curve, its linear sensitivity analysis, and the matched standardized models agree on the positive direction of the competition association, at a magnitude estimated in the 6–15 pp range across specifications. The empirical P75-versus-P25 estimate is a summary of one continuously estimated dose–response curve, not evidence for a threshold or for share as the clinically privileged scale. Likewise, the exact 25%, 50%, and 75% values shown on the profile-specific curve are interpretive anchors on that same curve: no athlete groups were formed at those values, and the model did not estimate change-points at 25%, 50%, or 75%. The profile-specific outputs add clinical interpretability by expressing fitted associations as absolute injury probabilities while holding mea-

sured characteristics constant; the same models can generate predictions for other specified profiles, and validating them as individual risk calculators is a task for future, externally validated work. The appropriately scoped conclusion is that greater reported competition-period carbon exposure among users was associated with greater adjusted injury prevalence on both examined scales—a population-level signal worth acting on through monitoring, even though its precise magnitude is not yet settled. Prospective measurement of unique carbon-shoe sessions, duration, intensity, and temporal order is needed to distinguish relative from cumulative exposure and to assess causality.

The OSTRC findings also show why injury burden should not be reduced to absence or complete cessation from sport. Time-loss-only systems can underestimate overuse problems that impair training, performance, or symptoms while athletes continue to participate [Clarsen et al., 2013, 2020]. Direct athlete reporting therefore complements injury counts and time loss with consequences that matter to the athlete. In this study the adapted score is best understood as a sport-specific athlete-reported outcome measure, not as a generic quality-of-life instrument or a substitute for prospectively repeated OSTRC surveillance.

4.3 Comparison with prior evidence

This study adds a population-level description to a literature that has so far relied on biomechanical and case-based work. A kinetic comparison of traditional and carbon-plated sprint spikes reported higher ankle joint contact force and peak plantarflexor force in the carbon-plated condition, with substantial inter-individual variability [Gerlach et al., 2025], providing a mechanistic account well aligned with the dose-sensitive pattern reported here. Case-based literature on Achilles tendon load with carbon-plate use [Tenforde et al., 2023] points in the same direction as the tendinopathy pattern reported here: tendinopathy was both over-represented as an injury type relative to the typical athletics pattern [Edouard et al., 2020] and disproportionately located at the foot, lower leg, and ankle (Table 9), consistent with that case-based signal at the population level for the first time. The same case-based literature has also flagged navicular and cuneiform bone stress with carbon-plate use [Tenforde et al., 2023]; this cohort’s elevated overall rate of bone stress injury is compatible with that concern in aggregate, but the injuries were thigh-dominant rather than foot-dominant in location, and the survey did not resolve bone-specific site (for example, navicular versus tibial), so this cohort’s data support the tendinopathy hypothesis more directly than the site-specific bone-stress-injury hypothesis. To the authors’ knowledge, this is among the first studies to describe carbon-plate insole exposure and its association with injury prevalence at the population level in a national sample, extending this prior mechanistic and case-based work with cohort-level evidence; the tendinopathy and bone-stress-injury patterns observed here are hypothesis-generating findings that a dedicated, adequately powered type-stratified analysis should test directly.

4.4 Strengths and considerations for interpretation

The two-phase recruitment, national age–sex poststratification, and Bayesian multilevel design were built specifically to support a defensible population-level estimate from survey data, and several features of the results reflect that: consistent direction across sensitivity specifications, well-behaved sampling diagnostics, and posterior predictive checks that closely reproduced observed injury counts. A number of features of the data are useful to keep in mind when applying these results. Because exposure and outcome were recalled together for the same season, this cross-sectional design describes an association rather than establishing temporal order between carbon-exposure increase and injury onset; a prospective design with objectively measured exposure is the natural next step. The survey’s timing compounds this: administered near the end of the season, it asked athletes to recall the

competition period from recent memory but the preparation period from further back, a differential recall window that could itself help explain why the preparation-period association came through weaker and less certain than the competition-period one (Section 2.11). Both exposure scales were derived from ordinal weekly training-type-frequency items. In particular, the absolute score is not an exact count of unique sessions: responses were top-coded and training domains may overlap within a session. The scale comparison was added to clarify interpretation after the primary analysis and should be regarded as secondary. The competition-period non-user group, while smaller (44 of 352 athletes) than the user group, reflects the near-universal adoption of carbon-plate spikes in this population and widens the credible interval for the adoption contrast. The association magnitude varied with functional form and hyperprior width, so the primary HSGP estimate of 13.2 pp is evidence of a positive association of plausibly meaningful size, with a magnitude this study’s data place somewhere in a 6–15 pp range. The adjustment set covered principal risk factors identified in prior running-injury literature [Taunton et al., 2003]; coaching-driven equipment allocation, session intensity, and baseline plantarflexor strength were not measured. Self-report enabled national-scale surveillance [Bahr et al.], but objective training logs are needed to validate the exposure measures. The adapted OSTRC-H2 items were completed by 94.6% of injured athletes but were asked only once and only after injury was reported, so they cannot recover weekly prevalence, symptom duration, cumulative burden, or scores among athletes who did not classify themselves as injured; the period-stratified SMDs are correspondingly descriptive. The study protocol was prospectively reviewed and approved by the Comitato Etico of the University of Turin (Section 2.2), providing a documented, ethics-verified record of pre-specification alongside this manuscript.

4.5 Generalizability

The two-phase recruitment and age–sex poststratification were designed specifically to support generalizability to the national competitive population; the modest Kish effective-sample-size retention after weighting (97%) and the close agreement between weighted and unweighted estimates (Appendix Table 19) support good generalizability to Italian FIDAL-affiliated competitive athletes across the age categories studied. Extending these findings to recreational athletes, road-running populations, or national contexts with different equipment-access patterns is a promising direction for future work, particularly given the near-universal adoption of carbon-plate spikes observed in this competitive cohort.

4.6 Conclusion

In this national cohort of competitive Italian athletics athletes, carbon-plated spikes were the prevailing training equipment in both preparation and competition. Adoption was not clearly associated with injury prevalence. Among competition-period users, greater reported carbon exposure was associated with higher adjusted injury prevalence whether represented as share or absolute frequency score; preparation-period estimates remained uncertain, and the data did not identify one exposure scale as superior. These findings support prospective surveillance using objectively measured unique sessions, duration, intensity, and known temporal order. Until such evidence is available, athletes, coaches, and clinicians should interpret the observed association as a signal for monitoring rather than as a causal risk estimate or a validated threshold for limiting carbon-plate use.

Ethics

This study was approved by the Comitato Etico of the University of Turin (protocol n. 0453001, approved 23 June 2026). All athletes provided electronic informed consent before completing the survey; athletes under the age of majority provided assent with parental/guardian consent as required by the approved protocol. Data were pseudonymized at the point of collection and analyzed under a study identifier; no directly identifying information (name, contact details, club affiliation) is reported in this manuscript or in any derived table or figure, in line with the data-privacy provisions of the Declaration of Helsinki and with STROBE-SIIS item SIIS 23.1 [Bahr et al.]. The two-phase recruitment and poststratification procedures described in Section 2.4 and Section 2.9 were part of the ethics-approved protocol.

Funding

None declared.

Data and code availability

Analysis code, aggregate tables, and the reproducible manuscript source are publicly available at <https://github.com/MVittu/ATLETA>. Individual-level survey responses and detailed recruitment-monitoring cells are not publicly shared because participant consent and ethics safeguards do not authorize open release.

References

- Roald Bahr, Benjamin Clarsen, Wayne Derman, Jiri Dvorak, Carolyn A. Emery, Caroline F. Finch, Martin Hagglund, Astrid Junge, Simon Kemp, Karim M. Khan, Stephen W. Marshall, Willem Meeuwisse, Margo Mountjoy, John W. Orchard, Babette Pluim, Ken L. Quarrie, Bruce Reider, Martin Schwellnus, Torbjorn Soligard, Keith A. Stokes, Toomas Timpka, Evert Verhagen, Abhinav Bindra, Richard Budgett, Lars Engebretsen, Ugur Erdener, and Karim Chamari. International olympic committee consensus statement: Methods for recording and reporting of epidemiological data on injury and illness in sport 2020 (including the STROBE extension for sports injury and illness surveillance (STROBE-SIIS).
- Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017. doi: 10.18637/jss.v076.i01.
- Benjamin Clarsen, Grethe Myklebust, and Roald Bahr. Development and validation of a new method for the registration of overuse injuries in sports injury epidemiology: The Oslo Sports Trauma Research Centre (OSTRC) overuse injury questionnaire. *British Journal of Sports Medicine*, 47(8):495–502, 2013. doi: 10.1136/bjsports-2012-091524.
- Benjamin Clarsen, Ola Rønsen, Grethe Myklebust, Tonje Wåle Flørenes, and Roald Bahr. The Oslo Sports Trauma Research Center questionnaire on health problems: A new approach to prospective monitoring of illness and injury in elite athletes. *British Journal of Sports Medicine*, 48(9):754–760, 2014. doi: 10.1136/bjsports-2012-092087.

- Benjamin Clarsen, Roald Bahr, Grethe Myklebust, Stig Haugsbø Andersson, Sean Iain Docking, Michael Drew, Caroline F. Finch, Lauren Victoria Fortington, Joar Harøy, Karim M. Khan, Bill Moreau, Isabel S. Moore, Merete Møller, Dustin Nabhan, Rasmus Oestergaard Nielsen, Kati Pasanen, Martin Schwellnus, Torbjørn Soligard, and Evert Verhagen. Improved reporting of overuse injuries and health problems in sport: An update of the Oslo Sport Trauma Research Center questionnaires. *British Journal of Sports Medicine*, 54(7):390–396, 2020. doi: 10.1136/bjsports-2019-101337.
- Pascal Edouard, Laurent Navarro, Pedro Branco, Vincent Gremeaux, Toomas Timpka, and Astrid Junge. Injury frequency and characteristics (location, type, cause and severity) differed significantly among athletics (‘track and field’) disciplines during 14 international championships (2007–2018): Implications for medical service planning. *British Journal of Sports Medicine*, 54(3):159–167, 2020. doi: 10.1136/bjsports-2019-100717.
- Andrew Gelman. Struggles with survey weighting and regression modeling. *Statistical Science*, 22(2):153–164, 2007. doi: 10.1214/088342306000000691.
- Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, Boca Raton, FL, 3rd edition, 2013.
- Marius Gerlach, Marie-Louise de Graaf, Kim J. Boström, and Heiko Wagner. Effect of advanced footwear technology spikes on lower limb kinetics and sprint performance. *Footwear Science*, 17(2):69–77, 2025. doi: 10.1080/19424280.2025.2458322.
- Paul A. Harris, Robert Taylor, Robert Thielke, Jonathon Payne, Nathaniel Gonzalez, and Jose G. Conde. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *Journal of Biomedical Informatics*, 42(2):377–381, 2009. doi: 10.1016/j.jbi.2008.08.010.
- Paul A. Harris, Robert Taylor, Brenda L. Minor, Veida Elliott, Michelle Fernandez, Lindsay O’Neal, Laura McLeod, Giovanni Delacqua, Francesco Delacqua, Jacqueline Kirby, and Stephany N. Duda. The REDCap consortium: Building an international community of software platform partners. *Journal of Biomedical Informatics*, 95:103208, 2019. doi: 10.1016/j.jbi.2019.103208.
- Eleanor Hayes-Larson, Katrina L. Kezios, Stephen J. Mooney, and Gina Lovasi. Who is in this study, anyway? guidelines for a useful Table 1. *Journal of Clinical Epidemiology*, 114:125–132, 2019. doi: 10.1016/j.jclinepi.2019.06.011.
- Laura A. Healey, Mathieu Bertschy, Shalaya Kipp, and Wouter Hoogkamer. Can we quantify the benefits of “super spikes” in track running? *Sports Medicine*, 52(6):1211–1218, 2022. doi: 10.1007/s40279-022-01657-4.
- Daniel Martínez-Silván, Eirik H. Wik, Juan M. Alonso, Elodie Jeanguyot, Bojana Salcinovic, and Andrew Johnson. Injury characteristics in male youth athletics: A five-season prospective study in a full-time sports academy. *British Journal of Sports Medicine*, 55(17):954–960, 2021. doi: 10.1136/bjsports-2020-102373.
- Joel Mason, Dominik Niedziela, Jean-Benoît Morin, Andreas Groll, and Astrid Zech. The potential impact of advanced footwear technology on the recent evolution of elite sprint performances. *PeerJ*, 11:e16433, 2023. doi: 10.7717/peerj.16433.

- Alannah K. A. McKay, Trent Stellingwerff, Ella S. Smith, David T. Martin, Iñigo Mujika, Victoria L. Goosey-Tolfrey, Jeremy Sheppard, and Louise M. Burke. Defining training and performance caliber: A participant classification framework. *International Journal of Sports Physiology and Performance*, 17(2):317–331, 2022. doi: 10.1123/ijsp.2021-0451.
- John M. Neuhaus, John D. Kalbfleisch, and Walter W. Hauck. A comparison of cluster-specific and population-averaged approaches for analyzing correlated binary data. *International Statistical Review*, 59(1):25–35, 1991. doi: 10.2307/1403572.
- Gabriel Riutort-Mayol, Paul-Christian Bürkner, Michael R. Andersen, Arno Solin, and Aki Vehtari. Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming. *Statistics and Computing*, 33(1):17, 2023. doi: 10.1007/s11222-022-10167-2.
- Jack E. Taunton, Michael B. Ryan, Douglas B. Clement, Donald C. McKenzie, David R. Lloyd-Smith, and Bruno D. Zumbo. A prospective study of running injuries: The Vancouver Sun Run “in training” clinics. *British Journal of Sports Medicine*, 37(3):239–244, 2003. doi: 10.1136/bjsm.37.3.239.
- Adam S. Tenforde, Tim Hoenig, Amol Saxena, and Karsten Hollander. Bone stress injuries in runners using carbon fiber plate footwear. *Sports Medicine*, 53(8):1499–1505, 2023. doi: 10.1007/s40279-023-01818-z.
- Toomas Timpka, Juan-Manuel Alonso, Jenny Jacobsson, Astrid Junge, Pedro Branco, Benjamin Clarsen, Jan Kowalski, Margo Mountjoy, Sverker Nilsson, Babette Pluim, Per Renström, Ola Ronsén, Kathrin Steffen, and Pascal Edouard. Injury and illness definitions and data collection procedures for use in epidemiological studies in athletics (track and field): Consensus statement. *British Journal of Sports Medicine*, 48(7):483–490, 2014. doi: 10.1136/bjsports-2013-093241.
- Toomas Timpka, Jenny Jacobsson, Joakim Ekberg, Caroline F. Finch, Jerome Bichenbach, Pascal Edouard, Michael F. Bergeron, Willie O. Roberts, Juan-Manuel Alonso, and Per A. Renström. Meta-narrative analysis of sports injury reporting practices based on the injury definitions concept framework (IDCF): A review of consensus statements and epidemiological studies in athletics (track and field). *Journal of Science and Medicine in Sport*, 18(6):643–650, 2015. doi: 10.1016/j.jsams.2014.11.393.
- Aki Vehtari, Andrew Gelman, and Jonah Gabry. Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, 27(5):1413–1432, 2017. doi: 10.1007/s11222-016-9696-4.
- Erik von Elm, Douglas G. Altman, Matthias Egger, Stuart J. Pocock, Peter C. Gøtzsche, and Jan P. Vandenbroucke. The strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *The Lancet*, 370(9596):1453–1457, 2007. doi: 10.1016/S0140-6736(07)61602-X.
- Steffen Willwacher, Manuel Kurz, Christian Menne, Elmar Schrödter, and Gert-Peter Brüggemann. Biomechanical response to altered footwear longitudinal bending stiffness in the early acceleration phase of sprinting. *Footwear Science*, 8(2):99–108, 2016. doi: 10.1080/19424280.2016.1144653.

A STROBE-SIIS reporting checklist

Table 12 follows the STROBE Sports Injury and Illness Surveillance (STROBE-SIIS) checklist [Bahr et al.], itself an extension of the STROBE statement for observational studies [von Elm et al., 2007]. The final column gives the manuscript page reporting each item, resolved automatically at compilation so that it remains accurate under re-typesetting; line numbers are printed in the margin throughout the manuscript (via `lineno`) to allow more precise location within the cited page.

Table 12: STROBE-SIIS reporting checklist, with the manuscript page reporting each item.

Item	Recommendation	Section	Page
1a	Study design term in title/abstract	Title	p. 1
1b, SIIS1.1, SIIS1.2	Informative abstract; sport/population/level; duration of observation	Abstract	p. 1
2	Background and rationale	Introduction	2
3, SIIS3.2	Objectives and specific purpose	Introduction	2
SIIS3.1	Registration status	Study registration	3
4, SIIS4.4	Study design; athlete/sport categorization	Design; Covariates	3, 6
SIIS4.1	Health problems observed	Outcome	5
SIIS4.2, SIIS4.3	Recording approach and coding	Exposure; Outcome	5, 5
5, SIIS5.1–5.3	Setting, location, dates, prospective/retrospective	Setting; Participants	4, 4
6	Eligibility, selection, recruitment	Participants	4
7, SIIS7.1, SIIS7.2	Variable definitions; outcome identification	Exposure; Outcome	5, 5
8, SIIS8.1–SIIS8.5	Data sources and measurement; timing; duration	Setting; Bias	4, 10
9, SIIS9.1, SIIS9.2	Bias and validation/reliability	Bias	10
10	Study size derivation	Participants	4
11, SIIS11.1, SIIS11.2	Quantitative variables; multiple-problem handling; severity	Outcome; Exposure	5, 5
12a	Statistical methods	Statistics	6
12b	Subgroup/interaction methods	Subgroups	10
12c	Missing-data handling	Missing data	11
12d	Sampling-strategy analytic accounting	Weighting	10
12e	Sensitivity analyses	Validation	10
SIIS12.1	Exposure-to-risk adjustment	Statistics	6
SIIS12.2	Risk-measure calculation	Estimands	7
SIIS12.4	Multiple health-problem handling	Outcome	5
SIIS12.5	Handling of athletes not included at outset	Missing data	11
SIIS12.6	Follow-up/censoring management	Weighting	10
13, SIIS13.1–SIIS13.3	Participant flow and reasons for exclusion	Sample flow	11
14, SIIS14.1	Descriptive characteristics by level of competition	Baseline characteristics	11
15, SIIS15.1	Outcome data; athletes vs. problems distinction	Descriptive use; Primary results	12, 15
16, SIIS16.1, SIIS16.2	Main adjusted results with precision	Primary results	15

(continued on next page)

(continued)

Item	Recommendation	Section	Page
17, SIIS17.1	Other analyses; injury region/type in tab- ular form	Injury pat- tern; Sensitivity; Weighted/subgroup	13, 16, 17
18	Key results summary	Discussion	27
19, SIIS19.1	Limitations	Limitations	28
20	Interpretation	Interpretation	27
21, SIIS21.1	Generalizability	Generalizability	29
22	Funding	Funding	30
23, SIIS23.1	Ethics and data privacy	Ethics	30

B Software and session information

All statistical analyses were conducted in R. Table 13 lists the software, model family, and sampling configuration used for the primary Bayesian multilevel model and its validation; runs used base seed 20260719 with deterministic model-specific offsets for reproducibility.

Table 13: Software, model, and posterior-sampling configuration.

Component	Version
R	4.5.2 (2025-10-31 ucrt), R Core Team
Platform toolchain	Rtools 4.5 (x86_64-w64-mingw32)
rstan	2.32.7
StanHeaders	2.32.10
loo	2.8.0
splines	4.5.2 (base R)
Model family	Bernoulli likelihood with logit link
Sampler	Stan Hamiltonian Monte Carlo, No-U-Turn Sampler
Chains $\times$ iterations	4 $\times$ 2000 per fitted model
Warm-up/adaptation	1000 iterations per chain (discarded)
Retained posterior draws	1000 per chain; 4000 per fitted model
Target acceptance probability	<code>adapt_delta=0.98</code>
Maximum tree depth	12
Random seed	20260719 plus deterministic model-specific offsets

Model code (`model.stan`, `weighted_model.stan`), the full analysis pipeline (`analysis.R`, `population_analysis.R`), and every derived table and figure referenced in this manuscript are retained under `experiments/multilevel/` in the project repository. The matched relative-versus-absolute exposure models are fitted by `fit_exposure_comparison.R`. The revised comparison, adoption-density, continuous-share, and descriptive sex-palette figures are generated by the corresponding `plot_*.R` scripts. A plain-language walkthrough of the complete analysis sequence is provided in `paper/latex/analyses-flow.md`.

C Supplementary tables

Table 14: Discipline support in the balanced analysis cohort. Codes 8–11 (relay, throws, race walk, and one further unused code) had zero respondents and were excluded from discipline-level adjustment.

Code	Discipline	n	Included in model
1	Sprint	107	Yes
2	Middle distance	69	Yes
3	Long distance	59	Yes
4	Hurdles	96	Yes
5	Horizontal jumps	44	Yes
6	Pole vault	19	Yes
7	Combined events	41	Yes
8–11	(zero respondents)	0	No

Table 15: FIDAL competition-registry counts used to construct the 2026 eligible-population frame. Under 20 counts for 2026 were used directly. Because 2026 counts were not yet available for Under 23 and Senior athletes when recruitment was planned, their sex-specific frame estimates were the rounded means of the 2023–2025 counts. The resulting age–sex frame totaled 1328 athletes; 2025 discipline shares within each age–sex cell were used to allocate stratum targets.

Age group	Sex	2023	2024	2025	2026	Frame estimate
Under 20	Women	235	265	274	204	204
Under 20	Men	327	361	338	311	311
Under 23	Women	217	211	207	–	212
Under 23	Men	244	247	252	–	248
Senior	Women	176	176	156	–	169
Senior	Men	195	199	159	–	184
Total		1394	1459	1386	–	1328

Table 16: Bayesian worst-case population-proportion precision calculation used to set Phase 2 (targeted recall) enrollment targets. The calculation set a minimum recruitment target; representativeness was addressed separately through stratified recall and poststratification.

Quantity	Value
Planning quantity	Population proportion
Prior	Beta(1,1)
Worst-case count	$x = \lfloor n/2 \rfloor$
Posterior at candidate n	Beta($1 + x, 1 + n - x$)
Interval and target	Equal-tailed 95% CrI; half-width ≤ 0.06
$n = 263$ half-width	0.060038
$n = 264$ half-width	0.059926
Estimated eligible population	1328
Minimum target sample	264 (19.9% of frame)
Consented, eligible, complete	355 (26.7% of frame)
Balanced analysis cohort	352 (26.5% of frame)
Target allocation	Proportional across 42 strata

Table 17: Age–sex poststratification weight diagnostics (Section 2.9). Model weight is the regularized ratio of target population share to sample share; Kish effective sample size after weighting was 340.5 (97% of $N = 352$).

Age group	Sex	Target n	Sample n	Sample share	Model weight
Under 20	Women	204	43	12.2%	1.26
Under 23	Women	212	53	15.1%	1.06
Senior	Women	169	62	17.6%	0.72
Under 20	Men	311	71	20.2%	1.16
Under 23	Men	248	63	17.9%	1.04
Senior	Men	184	60	17.0%	0.81

Table 18: Age- and sex-conditional posterior estimate of the competition-period dose association (75th vs. 25th percentile carbon share among users) from the primary HSGP model, holding the model structure and covariate adjustment fixed. P_{dir} is the posterior probability of the median estimate’s direction. These are descriptive, hypothesis-generating comparisons (Section 2.10), not independently powered tests of effect modification.

Grouping	Level	Difference, pp (95% CrI)	P_{dir}
Sex	Men	14.0 (−0.2 to 30.6)	96.8%
Sex	Women	13.1 (−0.1 to 29.1)	96.8%
Age group	Under 20	13.1 (−0.1 to 28.8)	96.8%
Age group	Under 23	14.0 (−0.2 to 30.6)	96.8%
Age group	Senior	14.1 (−0.2 to 30.8)	96.8%

Table 19: Population-weighted versus sample (unweighted) adjusted estimands from the primary HSGP model. Weighting applies the age–sex poststratification of Table 17. P_{dir} is reported beside every 95% CrI.

Period	Contrast	Unweighted, pp (95% CrI)	P_{dir}	Weighted, pp (95% CrI)	P_{dir}
Preparation	Adoption difference	6.5 (−5.6 to 19.2)	85.3%	4.8 (−7.7 to 16.8)	77.6%
Preparation	Dose difference	0.0 (−12.0 to 10.2)	50.6%	0.3 (−10.4 to 10.4)	54.5%
Competition	Adoption difference	4.7 (−9.6 to 18.6)	74.7%	7.1 (−7.4 to 20.6)	84.0%
Competition	Dose difference	13.2 (−0.2 to 29.1)	96.9%	13.7 (−0.2 to 30.0)	96.8%

Table 20: Model comparison and validation diagnostics for the primary HSGP (exposure) and matched baseline (no-exposure) models. ELPD: expected log predictive density (athlete-grouped PSIS-LOO); all Pareto- $k \leq 0.5$ for both models.

Model	ELPD _{LOO}	SE	Max $\hat{R}$	Divergences
Primary (exposure)	−429.00	11.14	1.005	0
Baseline (no exposure)	−430.92	11.03	1.007	0
ELPD difference (exposure − baseline): 1.92 (SE 2.41)				